When a Leaderboard Cannot Be Read: Three Rulers for the Noise Floor of LLM-as-Judge Evaluation

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Abstract

When a benchmark's labels come from language models, the label producers could compete in the ranking they defined. We treat a gold generator's leave-one-out score as a *noise floor*: under a plurality convention it is the maximum over generators. The audit emits a readable verdict only when the leader clears that floor by a stated margin; this is the positive branch of one producer-baseline rule, not a claim that every resolvable ranking must pass it. The floor belongs to how the gold was defined, so it must be measured before a ranking is cited under this rule. No public board measured here clears that rule: the contribution is the refusal semantics, the three rulers, and the measurements that force the refusal, not a demonstration that a public ranking is readable.

Three rulers are implemented (leave-one-out over model gold; a median-annotator substitution for human gold; a latent-variable Dawid–Skene arm that estimates a different quantity and feeds no verdict), separating refusals from graded readings: a corpus whose premises fail returns no verdict. Both public corpora have two generators, and under either tie rule their item sets are non-coextensive, so the aligning rule reduces the floor to an agreement rate. A 25-generator human-gold construction does return verdicts: three unreadable readings, each with the leader at its own floor, plus one domain refusal, and a released panel of 17 judge models returns the same zero margin. Where the ranked field is the pool that made the gold, the declared vacate-on-tie convention makes the plurality floor the leader's own score, so the margin has nothing to bind: the zero is an identity. Human gold leaves the floor undefined, and the latent arm measures no task.

No public board measured here receives a readable verdict from the rule as declared, and a calibration on a released panel certifies only its own debiased alignment: 1,242 of 9,568 resampled boards, against none uncorrected, on a pool whose contestants are its judges. A preregistered disjoint-field deployment on a second release, six judges and five contestants sharing no members, certifies none of 3,200 boards — the disjoint case is measured, and it measures a refusal at a negative true margin. We release the tool and its test suite.

1 Introduction

Leaderboards built on model-based judges are now common, and their conclusions are quoted as if the labels were ground truth. They are not. When the gold labels for a task are obtained by asking one or more language models to judge outputs, the models that produced the labels occupy a position that the framing of "contestant" does not describe: they helped choose the yardstick. Nothing prevents such a model from scoring well on the benchmark it helped define. This is not a hypothetical concern, and it is not a new claim. That judges disagree with people at non-trivial rates, and that agreement between judges is not agreement with truth, is established work rather than a new claim here; §3 states what that literature does and does not settle. Our claim is narrower and easier to act on. We do not estimate judge error rates and propagate them; we ask one question of a finished leaderboard: *does the leader clear the score that the gold generator*

would have achieved? If the answer is no, the leader cannot be shown to exceed the process that produced the gold, and this instrument does not certify the ordering above that level, however large the reported gaps look. It does not follow that the board carries no information: a refusal is a statement about what this ruler can certify, not a claim that the systems are indistinguishable.

What the rule certifies. The rule below answers one question: does the leader's score exceed the score of the process that produced the gold? It does not estimate a ranking error rate, and it does not certify that two systems are distinguishable. The perfect-gold case of §4.5 makes the difference concrete: with a 100% floor, a board whose top two systems sit at 90% and 50% is refused even though that ordering is plainly resolvable. A readable verdict is therefore a property of this audit's positive branch: the leader clears the producer baseline by the declared margin. It is not a necessary condition for every resolvable ranking, because the perfect-gold case above can refuse a plainly ordered 90% versus 50% comparison, and it is not sufficient to establish that the ordering below the leader is valid. The quantity is not new; the framing is. Classical test theory has long separated an observed score from the true score plus error, and the reliability of an instrument is reported through coefficients such as Cronbach's coefficient (9), Krippendorff's coefficient for multiple coders of categorical data (21), and the intraclass correlation coefficient for agreement between raters (31, 20). We use the last of these, and report it in the absolute-agreement form, because the question we ask is whether two measurements can be substituted for one another rather than whether they rank in the same order. The same literature supplies the correction we needed for our own comparisons: observations clustered within cases cannot be treated as independent in a bootstrap, which is why we report intervals under both the item and the case as the unit (5). We call the quantity the *noise floor*: the score obtained by the gold-generating process itself, when scored against its own output under a convention that removes its own votes from the gold. It is a floor for this audit in the narrow sense that a readable verdict requires the leader to exceed it under the declared margin; the quantity does not by itself measure ranking error or certify that every lower ordering is resolvable. Removing its own votes rather than keeping them is what keeps the quantity from being self-fulfilling: a generator scored against gold containing its own answers is credited for them. Three design choices make the floor non-obvious, and each of them changes the verdict. First, the gold definition: majority vote and mean-probability voting disagree on which systems lead. Second, the aggregation: macro (average over items) and micro (pooled accuracy) differ by up to 6.75 points per system under the tie rule that resolves a disagreement by a fixed order, and by up to 5.02 points under the rule that leaves it unmeasured (§5.3). Third, the reference itself: a latent-variable model of the same annotations yields a different floor again. A paper that reports a verdict without naming these choices is reporting a number whose meaning has been decided by unstated defaults. Two further choices are easier to miss because they are bookkeeping rather than modelling. The fourth is whether the floor and the leaderboard are computed on the same items: a difference between a floor measured on one denominator and a leaderboard measured on another is not a margin, so the tool refuses to report one (§5.1). The fifth is what happens when the gold generators cannot agree. A conventional vote has no output on such an item, and every implementation must then decide—leave it unlabelled, or break the tie by some rule—and that decision changes the size of the labelled set and therefore whether a verdict exists at all. We treat it as a declared parameter: the tool exposes it as `--tie-rule`, refuses to use the forced variant for a verdict, and reports both readings side by side (§5.1). We make four contributions. (1) A definition of the noise floor together with a decision rule on it, with the margin stated once as a declared constant and printed beside every margin, so that a reader can re-threshold without recomputation. (2) An implementation providing three rulers: leave-one-out under judge-generated gold, human gold, and a Dawid–Skene latent model, including an explicit refusal when the annotations fail to identify the quantities being estimated. (3) Measurements on two public corpora showing that the verdict is determined by design choices, and quantifying how much of the ranking is design-determined. (4) A validation methodology: we treat our own test suite as an instrument that can be wrong, and we release the mutation battery that measures how much of it can fail.

2 Threat Model

A noise floor is only meaningful once "the gold generator" is defined. Three situations, in increasing order of how much they threaten the construction:

What is being threatened. A ranked table whose gold labels were produced by some process invites a stronger reading than the table itself records: that the top entry's advantage is meaningful relative to the process that made the gold, while the table usually does not state the reproducibility of that process. The threat modelled here is therefore not misconduct but an underdetermined reading: the conventions a benchmark does not name can decide whether this audit returns a positive branch, a refusal, or no verdict at all. This audit tests whether the leader clears one named producer baseline; it does not establish the validity of the full ordering. We state the three cases according to who made the gold, then five invariants that the rulers of §4 rely on, each with the place in the tool that enforces it or reports its violation.

The gold is human. No model helped write the labels. A leave-one-out floor over models is undefined, because removing a model's votes leaves a gold that the model did not help construct. We handle this in §4.3 by refusing to compute the floor and substituting a stated reference.

The gold is a consensus of several models. Then every model in the consensus is a gold generator. The floor of each is computable under each convention, and the maximum over generators is the relevant bound, since a leader must clear all of them.

The gold is a single model. This is the worst case and the easiest to overlook, because a single-model judge produces a leaderboard that looks exactly like a human-annotated one. The quantity this rule would ask for is that model's own accuracy on the tasks it labelled, and under the inclusive reading of (1), which scores a generator against gold that contains its own answers, that accuracy is identically 1: the board would read as collapsed by the estimator rather than by the leaderboard. The tool does not go there. With $m = 1$ the leave-one-out reduction leaves no second process to reproduce, so the same guard that fires at $m = 2$ returns the floor-degenerate refusal under both tie rules, which is the behaviour §8 and the test suite pin (two cases, one per tie rule). We do not treat the human case as automatically safe. Human labels carry annotator disagreement, and a benchmark that reports no annotator agreement has not shown that its labels are reproducible. Our tool does not resolve that question; it reports the annotator-level reproduction rate when the data permits, and says so when it does not.

Invariants. Five conditions, each with what the tool does when it is violated. Two of them stop a verdict, two are disclosed alongside one, and one is a declared parameter; the distinction is ours to make explicit, because a gate that fired and a check that was not written leave the same reading on the page.

1. **Generator sufficiency, $m \geq 3$.** With fewer than three gold generators, removing one leaves a single generator, so (1) measures agreement between two processes rather than one process reproducing itself. The ruler returns UNDECIDABLE_FLOOR_DEGENERATE and reports the per-generator values it would have compared (Algorithm 1). The condition is necessary for the quantity to be defined, not sufficient for it to be unbiased: three generators that share training data or a labelling convention can reproduce one another's errors, and a third generator does not by itself guarantee that the remaining two avoid a tie. We treat $m \geq 3$ as an applicability condition, not as evidence of independence.
2. **Domain identity.** The floor and the leaderboard must be scored on the same items, and the gate tests membership rather than the two counts, since two sets of the same size with different members are as unusable a pair as two sets of different sizes and are the pair a count check passes. The test runs over *every* scored system, not only the leader: `domain_membership_ok()` compares each system's own item set against the floor's, and a single system measured elsewhere is enough to refuse the board, because a margin is read off a ranking whose rows are supposed to share a denominator. The offenders are named in `domain_membership_off_systems`. Where membership differs the ruler returns UNDECIDABLE_DOMAIN_MISMATCH, because the terms were measured over different item sets and their difference is not an estimate of anything; §5.1 is the case where this gate, rather than the floor, is the finding.
3. **A declared tie convention.** When generators split, an implementation must either leave the item unlabelled or break the tie by a rule, and the choice changes the size of the labelled set. The tool exposes it as `--tie-rule`, refuses to use the forced variant for a verdict, and reports both readings side by side.

4. **Disclosed actor overlap.** Where the gold generators are also contestants, the floor is constructed by the graded. The auditor computes that overlap on every report (`self_grading_check`), marks the resulting floor optimistic when the overlap is non-empty, and returns `checkable` false rather than a guess when a table carries no actor column. The condition is disclosed rather than gated, and §5.4 measures what it costs.

5. **Resolution at the declared margin.** δ in (3) is a convention, so a design can be unable to resolve a difference of that size. The tool then reports the median minimum detectable difference and the approximate power at δ (`mde_median`, `power_at_delta`) beside the margin it fails to resolve, and §5.4 gives four tasks where the reported MDE exceeds δ.

3 Related Work

Measurement-theoretic limits of model-based evaluation have been taken up in several literatures, grouped below as seven. We map them to the invariants of §2 rather than to a taxonomy of papers; one of the five, the declared tie convention, has no measurement-theoretic literature behind it and is a bookkeeping choice, so it is not represented below.

Common denominator. The requirement that two statistics share a measurement domain before their difference can carry a margin is our own invariant (§2) rather than a result we inherit; what (9) supplies is the reliability reading of a single instrument's output, not a statement about comparing two. (30) measures how a judge's verdict moves when the positions of the candidates move, and (34) reports that even its strongest judge falls short of the agreement humans reach with one another and that judge scores shift with the design of the judgement. Neither compares two gold definitions built from one annotation pool; the item-set divergence of §5.1 is ours, and we cite those two only for the adjacent claim that a judge-derived ranking is not a stable measurement.

Self-preference and self-recognition. (27) measures the mechanism by which a judge that helped write the gold can sit above a floor defined by it: with controlled generations, evaluators recognise and favour their own outputs. Our floor is not a correction for that mechanism and does not separate it from reproduction of the gold convention — on the 25/25 construction of §5.6 the two act in the same direction, which is why the section reports optimism as a stated property of the reading rather than as an estimated quantity. Where the generator pool and the ranked field are disjoint, the floor bounds alignment with the gold producers and says nothing about favouritism; the two claims should not be merged.

What a judge panel is measured against. (38) verifies an LLM judge against human preferences, and (33) scores judges on questions whose correct answer is human-verified precisely because crowd-sourced preference is a poor guide there. In both, the judge's own reliability is the object; neither derives a level that the ranked field must clear from the reproducibility of the judging pool. Our objection is therefore not to the number of judges on a panel but to what a two-generator pool can measure: at $m = 2$ leave-one-out leaves the other generator's answer, so the floor is an inter-generator agreement rate rather than a reproduction rate.

Panels of judges in the wild. A panel is standard practice rather than a proposal of ours: (35) replaces a single judge with a panel of diverse smaller models, and (38) scores its benchmark with model judges. What that literature reports is agreement among judges, not the floor a judge panel induces. Two published cases bracket the gap. (3) states its jury composition as GPT-4o, Claude 3.7 Sonnet and LLaMA 3.3 70B, and its public board lists nine evaluated models, three of which are those same jury members — the configuration that §5.6 measures, occurring in a maintained public suite rather than in a construction of ours. We did not recompute a floor for it, and we could not: its published table collapses the three jurors into a single aggregate, while the script that reads them per juror points at a private institutional path. The sentence we can defend is therefore structural rather than measured: once the ranked field includes the models that define the reference, floor and leaderboard stop being independent, and that is the condition under which our rule declines to read a verdict

at all. (24) builds the disjoint version of the design: seven judges, with the paper's own sentence that Llama models are excluded from the panel because they are the systems being simplified. Its per-judge vote table, which is what a floor would have to be recomputed from, is announced as a future release, and the public repository holds a README alone. So the phenomenon and the correct design both exist in the open literature, and one public artifact does carry the votes: the release `potsawee/truthful-qa-llm-judges` on the Hugging Face hub (retrieved 2026-10-13, and `evidence/hf_panel_release_probe.json` records its file list, README digest and retrieval time) puts one prediction file for each of 17 judge models over 5,918 items of the TruthfulQA suite (23), with the item's own benchmark label alongside each prediction. Its card names no paper, so we deliberately do not attribute it to the panel studies cited above; it is offered here only as evidence that the artifact our rule asks for — per-judge votes at $m \geq 3$, one file per judge — is publishable and has been published. The counts in that sentence are read from the hub's dataset index by `probe_hf_panel.py` and archived as `evidence/hf_panel_release_probe.json` (file list, README digest, retrieval time); the probe covers the one repository named above, and we did not verify any other release it may point at. The votes themselves are referenced, not redistributed, because that release declares no license. That is the input a floor would be recomputed from, and it is the piece (24)'s table is not yet. It is not an instance of the configuration this paper needs: the release ranks those same 17 models by their accuracy against the benchmark's own labels, so the pool that could define a model-generated gold and the field that would be ranked under it are again one and the same set of models, and the thing that keeps them independent is a human label rather than the panel. We searched the hub's dataset index (later through its API as well), arXiv full text, GitHub file trees, Kaggle's dataset index and OpenReview's public search; we did not search gated commercial boards. In the latest pass we found one release that meets the membership requirement: a three-judge panel scoring a single system, disjoint from its judges, with per-judge votes and an MIT licence (40); it publishes no ranked field, so it carries no leader comparison, and we name it as the closest artifact rather than as an instance. On the surfaces we did search we found no release with three or more model judges, downloadable per-judge votes, and a contestant pool disjoint from the judges. The panel release is the nearest thing to one, and §5.7 runs the ruler on it: it returns `UNREADABLE` with the floor equal to the leader's own score, at a margin of exactly zero, which is what Proposition 4 says must happen whenever a release ranks the models whose votes make its gold. The panel is therefore an instance of that identity rather than evidence about panels of this provenance, and it does not move the collapse of §5.6 from our construction onto somebody else's — the collapse sits in the shape, not in the assembly step. A disjoint-field case is measured in §5.9 and returns a refusal; what remains unmeasured there is a disjoint board that receives a positive verdict. §8 records that distinction as the scope limit of this paper.

Power-resolution bound. (5) and (1) give the design-effect correction behind $\text{deff} = 1 + (k-1)\text{ICC}$; the paired-difference resolution figure behind an unreported MDE is the ordinary two-sided normal calculation at the sample size actually resampled, which we state in §4.1 and §5.4 rather than attribute to a literature. (25) is the clearest statement of the neighbouring line that puts intervals and power calculations on benchmark scores by treating an evaluation as an experiment on a super-population. That is the sampling half of what we ask a report to carry, and it is a different quantity from ours: an interval can be narrow around a score that sits at or below the floor, in which case (3) reads `UNREADABLE` with a well-estimated margin. The ruler adds the level, not the bar.

Actor-disjointness. (10) models annotators as error-prone latent classes and (11) supplies the EM fit; (22) and (17) take annotator behaviour and label quality as their object. Neither literature studies a pool that is simultaneously the judge and the contestant: the 25/25 overlap is what §5.4 measures, and we claim it as a measurement of ours rather than a finding of theirs.

Inference stack as a third source of variation. The two literatures above locate variation in the judge and in the definition of the gold. A third layer sits underneath both and is not specific to model-based evaluation: the serving stack that turns a prompt into the tokens a verdict is computed from. (37) measures that layer directly, reporting that under bfloat16 precision and greedy decoding the accuracy of a reasoning model can differ by up to 9% and its response length by up to 9,000

tokens across configurations that differ only in GPU count, GPU type and evaluation batch size, and tracing it to the non-associativity of floating-point addition under limited precision, whose effect is amplified in reasoning models because a small rounding difference in an early token changes the chain of thought that follows. (13) reaches the same conclusion from the serving side and argues that the common "concurrency plus floating point" explanation is incomplete: a request's output depends on the batch it was scheduled into, because the reduction order inside a kernel changes with the batch size. A production framework treats this as a quantity to be handled, shipping a batch-invariant mode and a deterministic sampling mode, the latter requiring a fixed sampling seed and the speculative path disabled (39). The scheduling machinery of the same stack is documented separately: (7) derives the verification length of a speculative-decoding step from per-request prefix survival probabilities and an engine throughput profile, so even the work a request puts on the server depends on the load it happened to be under. We cite these four to bound where our floor can sit, not to measure it. Their quantity is a property of the serving configuration, and the item is the same question asked of a model and not of a judge reproducing a gold definition; none of them reports how much a judge's verdict moves when the stack underneath it is reconfigured, and §8 records the corresponding scope limit of this paper.

Two omissions are common to the seven groups above: a refusal semantics for the audit itself, which is ours and is not attributed above (the nearest classical statement is (21) on agreement under missing data), and a mutation-battery check that a test suite cannot pass by construction, likewise ours. We supply both, because the claim that a noise floor is measurable is a claim about what survived deletion, not about what was reported.

We inherit the prior art rather than reject it, but we change its burden of proof: where the literature above takes inter-rater agreement as a reliability statistic, we read it as a resolution bound and refuse to convert it into a ranking verdict.

3.1 What we add

The three literatures each supply a component, and our contribution is the measurement that requires all three at once. Judge-reliability work gives an error rate but not a verdict on a ranked list; classical test theory gives a reliability coefficient but not a decision rule; latent-variable models give a floor but require identifiability that two annotators do not provide. We supply the leave-one-out construction that turns a gold-generating process into a floor on the same items the leaderboard is scored on, the decision rule that consumes it with a declared threshold, and an explicit refusal in each of the three cases where the requested quantity has no referent: no gold generator to remove, fewer annotators than the model needs, or a floor and a leaderboard that do not share a denominator. We also treat our own test suite as an instrument that can be wrong and measure how much of it can fail (§6). A preregistered calibrated variant of the floor (§5.8) asks what the reading is worth when a ruler outside the ranked field exists to calibrate against, and answers on the released panel: certification becomes reachable there, its rate and its false-certification rate are both measured, and the scope — which configurations leave a margin to certify at all — is reported with the arm rather than assumed.

4 Method

4.1 The noise floor and the decision rule

Let a benchmark have n items, and let a set of gold-generating processes $G = \{g_1, \ldots, g_m\}$ produce reference labels. For a convention c defining how labels are combined from g, and for a system s, let $\mathrm{acc}_c(s)$ be s's accuracy against the gold produced under c. Define the leave-one-out floor

$$\mathrm{floor}_c(g) \;=\; \mathrm{acc}_{c^-}(g), \tag{1}$$

where c^- is the convention with g's own votes removed from the gold and $m-1$ generators remaining, and

$$\mathrm{floor}_c \;=\; \max_{g \in G} \mathrm{floor}_c(g), \tag{2}$$

because a leader must clear every generator, not the average of them. The leave-one-out construction is the one we report because an inclusive floor credits the generator with its own correct answers and therefore understates the noise; we report both. Three details of the construction are load-bearing rather than clerical. First, a floor computed on each generator's own item set and a leaderboard computed over all items subtract quantities with different denominators, so we compute both on the same item set—for the leave-one-out floor, the items that remain labelled after any single generator is removed—and compare the two sides as sets rather than as sizes: the emitted record carries each side's cardinality and a digest of its item set, and the gate fires on a difference of membership, since two equal-sized sets with different members are still a difference of denominators, and a difference of membership that a cardinality check does not see. Where the two admit no common set at all we do not repair the mismatch by choosing a different set: intersecting the floor's item set with the leaderboard's restricts the comparison to the items on which the generators already agreed, and on those items the leave-one-out value is degenerate by definition, so the floor would rise to the rate at which a generator agrees with itself. We report both sides' item sets and refuse. Second, a row whose item carries no gold label under the reduced convention has not been measured, so we exclude it from both the numerator and the denominator and report the coverage that this leaves; scoring it as wrong would report precision the corpus does not have. Where $m = 2$ the leave-one-out floor reduces to agreement between the two generators and is a weaker quantity than an independent reproduction rate; we compute it, label it `inter_generator_agreement` in the output, and record that $m \geq 3$ is required for the stronger reading. Third, when two generators vote differently the convention produces no label for that item and the labelled set shrinks. That shrinkage is a choice rather than a consequence: the tool takes it as its default, on the reading that an item two generators disagree about has not been measured rather than measured as half-right, and exposes adjudication by comparing the answer strings as a flag whose help text marks it as a sensitivity comparison that must not carry a verdict. We report both readings in §5.1, because on one of our two corpora the choice decides whether a verdict exists at all. A leaderboard is reported as readable when

$$\max_s \mathrm{acc}_c(s) \;>\; \mathrm{floor}_c + \delta, \qquad \text{where} \qquad \mathrm{floor}_c \overset{(2)}{=} \max_{g \in G} \mathrm{floor}_c(g), \qquad \delta = 2.0 \text{ points}, \tag{3}$$

so the rule has two parts: the leader must sit above the highest generator, and it must do so by more than the declared margin δ. The comparison is against the maximum over generators rather than their average, because a leader that clears the average of its gold producers has not cleared the one that agrees with it most. Two failure modes are named: UNREADABLE when the floor exceeds or equals the leader, and MARGINAL when the leader clears the floor by less than δ. One boundary is worth recording rather than absorbing: the formula is strict, while the implementation's branch is margin $< \delta$, so a margin of exactly δ reads as the readable branch. No reported margin sits at δ — the m25 and panel arms sit at 0.00 and the forced-rule decision margin is 2.96 — so no reading here depends on it; a reader re-running the tool should treat the equality case as marginal, which is what the formula requires. No reported margin sits at δ — the m25 and panel arms sit at 0.00 and the forced-rule decision margin is 2.96 — so no reading here depends on it; a reader re-running the tool should patch the comparison to $\leq$. The rule also has an operating condition, stated here rather than left to the limitations: for the comparison to be able to return a positive reading, the corpus has to resolve a difference of size δ, and on the four tasks where we can measure that resolution (§5.4) the median minimum detectable difference exceeds the declared threshold. The positive arm is therefore not resolvable at $\delta = 2.0$ for an effect of that size; a margin far above δ could still fire, but none of these corpora produces one; we report that as the rule's operating range rather than as a finding about them. One scope note belongs beside the rule rather than among the propositions: floor_c in (3) is (1) under the plurality convention, and no other aggregation is admitted into it. That distinction is not bookkeeping. §5.2 scores the same two generators under a probability-weighted convention in which removing a generator shifts an aggregate instead of vacating an item, and under that convention the system in first place changes; the deduction of Proposition 4 turns on the count, so it is offered for this rule only.

A refusal is not a verdict, and we keep the two in separate types throughout. The tool emits UNDECIDABLE_DOMAIN_MISMATCH when the floor and the leaderboard fail to share a denominator, and UNDECIDABLE_FLOOR_DEGENERATE when the gold has fewer than three generators and the leave-one-out quantity is therefore not the reproduction rate (1) was defined to be. Neither is a point on the UNREADABLE–MARGINAL–READABLE scale: what a refusal withholds is the verdict, not the readings. A refused comparison is still printed in full — floor, leader, their difference, and the paired interval

of that difference — and is marked as reference rather than as a basis for deciding, so a reader can see what was withheld and on which quantity the withheld verdict would have rested. Where the two terms sit on different item sets, that printed difference is a difference of two denominators, which is precisely why it carries no verdict. The decision-bearing domain of (3) is stated once here because it is narrow: three or more gold generators, and that pool a *strict* subset of the ranked field. Of the five readings this paper reports, the two $m = 2$ corpora fail the first condition and the $m = 25$ construction and the released panel fail the second, which is Proposition 4; the human-gold set has no model-generated gold at all. No board measured here produced a positive verdict from (3) about somebody else's leaderboard.

The threshold δ is a declared convention, not a derived quantity. That has a consequence we make explicit rather than leave implicit: a corpus can be unable to resolve a difference as small as δ, and on such a corpus the second part of (3) cannot bind. We therefore report, alongside each margin, the median minimum detectable difference (MDE) at the sample size actually used, and the approximate power at δ; where MDE exceeds δ we say so and report the paired interval beside the margin. The implementation does not gate on the interval, so a READABLE arm whose interval crosses zero would still print as READABLE; no data arm in this paper reaches that branch, and we do not read one as settled if it did. We report both numbers rather than choosing one, because a reader who prefers a different δ can apply it to the margins we give without recomputation. The same applies to the margin itself: every margin in §5.1 is reported with the interval obtained by resampling cases jointly for the floor and the leader, because a verdict read off a point estimate is a stronger claim than the data supports whenever that interval contains zero. We report margins rather than verdicts wherever the verdict could be read as a stronger claim than the margin supports. Note that a margin is only reportable once the floor and the leader share a denominator; where they do not, the correct output is a refusal, not a widened interval, and §5.1 is an instance in which the refusal is the finding.

4.1.1 Four structural facts, stated as propositions

The four constructions that decide most of what follows are worth stating separately, because each is a fact about our own rules rather than about any benchmark, and each can be checked against the implementation without re-running a measurement. The derivations are short in each case, and the point of the propositions is the statements rather than the deductions: a leaderboard audited with a broken estimator is unreadable for reasons that have nothing to do with the systems on it. Propositions 1–3 are about the two-generator corpora of §5.1 and §5.2; Proposition 4 is about every corpus in which the models that make the gold are also the models ranked by it, and it decides what the readings of §5.6 and §5.7 can mean.

Proposition 1 (the two-generator floor is an agreement rate). Let $m = 2$, and let $g \in \{g_1, g_2\}$ be the generator removed. Because the reduced convention c^- is fed only the remaining generator's vote on every item, the leave-one-out gold label is whichever label that generator casts; hence $\text{floor}_c(g) = \text{acc}_{c^-}(g) = \Pr[g = g']$, an identity about the rule rather than a published result. The quantity is in the unit of agreement between two annotators, not in the unit of accuracy against an external truth, and it is not the reproduction rate that (1) was defined to produce.

Proposition 2 (a two-generator mean-probability rule is a sign decision). Let the convention aggregate, for each answer a, by summing the probabilities its winners carry: $\text{acc}_c(a) = \sum_{i:\, c_i = a} p_i$, with $a^\star = \arg\max_a \text{acc}_c(a)$. This is the mean-probability branch of `build_gold()`. With $m = 2$ and two different labels, the two sums are p_1 and p_2, so $a^\star$ is the label of whichever generator holds the higher probability: a decision carried by the sign of a difference between two numbers, not by their size relative to any threshold, and unaffected by how much either generator believed the answer it cast. It converts a weak disagreement between two models into a gold label by consulting a second, unrelated detail of the generators, and so it leaves no item unmeasured. It follows from the definition of the aggregator and is not a statement about any particular corpus; on the decision corpus it is the reason the label adopted under the probability rule is the one carried by the more confident generator.

Why the identity matters. The one detail a disagreement could have supplied — the margin by which one generator defeated the other — is exactly the quantity the rule throws away. A gold built this way has the appearance of having resolved the disagreement and the substance of having flipped a coin: a high-confidence

label and a between-the-two-models tie both yield a confident winner, and the corpus appears fully measured. Our point is not that the rule mislabels particular items; it is that a rule which resolves every disagreement loses the distinction between a measured and an unmeasured item, so it is not a floor measured on all of the corpus. Under this rule a corpus does not shrink, a coverage statistic does not drop, and the shrinkage that would have told the reader "this is not being measured" is suppressed. We therefore expose the rule as a flag whose help text marks it as a sensitivity comparison that must not carry a verdict, and report both readings because on one of our two corpora the choice decides whether a verdict exists at all. Whether the rule is benign on any given corpus, and which of the two leaders it would promote there, are questions that only a corpus can settle; §5.2 measures both, and the measurement is what the verdict in §5.1 is made to depend on rather than the other way round.

Proposition 3 (at $m = 2$ aggregation is not recoverable from the labels alone). Let $m = 2$ and let a convention c be any function of the two votes. For a fixed set of vote pairs the gold under c is a function of the pair, so two conventions can be told apart on that corpus only by the labels they produce; where they produce the same labels on all items, no statistic computed from the labels can distinguish them. This is why the paper fixes the aggregation for every reported margin: a floor is only comparable to a leaderboard on the same aggregation.

Proposition 4 (where the ranked field is the voting pool, the floor *is* the leader). Let c be the plurality convention of (1): an item carries a label exactly when the largest vote count has a unique holder. Removing one generator g lowers exactly one answer's count by one and moves no other, so on any item that convention has a label for, the reduced convention either keeps *the same* label or meets a tie and keeps none. A leave-one-out reduction therefore never re-labels an item; we note what it does instead, which is stop being defined on some. Now score both sides on the one shared item set U that the domain gate of Algorithm 1 requires, take the ranked field to be the generating pool, and take each generator's ballot on an item to be its own answer to that item. Then for every generator $\mathrm{acc}_{c_g^-}(g)$ and $\mathrm{acc}_c(g)$ coincide on U, so

$$\mathrm{floor}_c \;=\; \max_{g \in G} \mathrm{acc}_{c_g^-}(g)\big|_U \;=\; \max_{s} \mathrm{acc}_c(s)\big|_U,$$

the floor equals the leader's own score, the margin of (3) is zero on any corpus of this shape at any sample size, and neither MARGINAL nor READABLE is reachable. We note the consequence: under these three hypotheses δ does not fail to bind because a leader is weak, it cannot bind.

What the proposition is for. It turns §5.6 and §5.7 from two measurements into one structural fact with two instances, and it names the only condition that makes a leave-one-out floor informative: the pool must be a *strict* subset of the ranked field, so that the leader can be a system no generator's ballot flatters. That is the design (24) states and has not released the votes for. Two of the three hypotheses are enforced rather than hoped for: the builder of the 25-generator table refuses to write it unless every actor's ballot equals its own answer on the same item, and prints that pairing as a receipt in `evidence/log_m25.txt`, and the domain gate refuses to compare a floor and a leaderboard that sit on different item sets. The third is a property of the data, and it is readable on disk: on every task of the construction that §5.6 measures, and for every generator in its pool, the number of items on which a reduced convention *disagrees in label* with the full one is zero (`label_change_pairs` in `evidence/m25_selfbias.json`, counted there task by task). The equality the two sections report is the conclusion of this deduction, not a coincidence at the top of a distribution. We also mark where the deduction stops, since its first step is a statement about a count and not about a score: under the probability-weighted convention of §5.2, removing a generator changes each answer's aggregate rather than vacating one item, so the holder of the largest aggregate can move and no such identity is available. That is not a gap we are papering over — §5.2 reports the two conventions disagreeing about which system leads, and Proposition 4 is deliberately not offered as a way to absorb that reading into itself.

Algorithm 1 is the protocol the two earlier sections describe in prose, written once as a specification so that a reader can check our implementation against the paper rather than against our description of it. Two properties are worth isolating from it. The first is that the refusals are *gates*, not scores: they return before any comparison with δ is made, which is why no choice of δ can rescue a corpus that fails one. The second is that the ruler is a decision procedure over quantities that are already computed. The human-gold ruler of

Algorithm 1 The leave-one-out ruler as implemented in `audit.py`, read off the decision chain in that file. The branch order is the order in the code, and that order is load-bearing: the domain gate runs before the degeneracy gate, so a corpus that fails both is reported as a domain mismatch. The three gates above the comparison with δ are refusals, and they are listed in the order the code tests them.

Require: item $\times$ generator vote matrix V; generator set G with $m = |G|$; tie rule; declared margin δ; bootstrap draws B
Ensure: one of the seven strings enumerated in §4.2

for $g \in G$ **do**
    $\text{floor}(g) \leftarrow$ accuracy of g against the gold of $G \setminus \{g\}$, on the items still labelled ▷ eq. (1)
end for
$\text{floor} \leftarrow \max_{g \in G} \text{floor}(g)$ ▷ eq. (2): a leader must clear every generator
$\Delta \leftarrow \text{lead} - \text{floor}$; $[\ell, u] \leftarrow$ the paired case-level bootstrap interval of Δ over the B draws, floor and leader resampled on the same rounds
$I_{\text{floor}} \leftarrow$ the item set supporting that floor; $I_s \leftarrow$ the item set actually scored for each system s on the board
if $I_s \neq I_{\text{floor}}$ for any s **then return** UNDECIDABLE_DOMAIN_MISMATCH ▷ one system measured elsewhere voids the ranking; equal sizes can hide one, and the offenders are named
else if $m < 3$ **then return** UNDECIDABLE_FLOOR_DEGENERATE ▷ leave-one-out degenerates to pairwise agreement, not self-reproduction
else if floor or lead is not a number **then return** UNDECIDABLE_NO_READING ▷ a run that measured nothing must not fall through to the best-looking arm
else if $\text{floor} \geq \text{lead}$ **then return** UNREADABLE ▷ includes equality: the declared rule is $\max_s \text{acc}_c(s) > \text{floor}_c + \delta$, so a zero margin clears nothing
else if $\text{lead} - \text{floor} < \delta$ **then return** MARGINAL
else
    return READABLE ▷ the paired interval of the margin is printed with it, not withheld
end if
Every branch above has already printed the floor, the leader, their difference and that difference's interval; a refusal prints them as reference and decides nothing on them.

§5.4 imposes the same requirement by a different mechanism—it restricts every actor to the items that all actors label, and when that intersection is empty it marks the pairwise comparisons incomparable and the median non-citable instead of ranking across denominators—and, having no leave-one-out floor to compare against, it reports the median annotator in its place.

4.2 The output alphabet, branch by branch

Algorithm 1 is the whole of the leave-one-out chain, but it is not the whole of what the tool can print, and the difference matters to anyone reading a released report. That chain emits six strings: the refusals `UNDECIDABLE_DOMAIN_MISMATCH`, `UNDECIDABLE_FLOOR_DEGENERATE` and `UNDECIDABLE_NO_READING`, and the graded readings `READABLE`, `MARGINAL`, `UNREADABLE`. Two of those six have no carrier among the arms this paper reports. `MARGINAL` is not returned by any of them, because the equality case falls to `UNREADABLE` at the equality branch, and no arm of §5.6 clears its floor by a positive amount; `UNDECIDABLE_NO_READING` is reached by no corpus either. Both are exercised by fixtures in `selftest.py` rather than by data, which is what keeps them from being decorative: the `bnf_floor_nan.csv` case exists because a run whose floor came back `nan` used to fall through both comparisons, since `nan` fails every inequality, and print `READABLE`.

The human-gold ruler of §5.4 has a second chain, and with it a separate alphabet. With no leave-one-out floor available it measures the field against the median annotator, and because that reference is not a floor it reports `ABOVE_MEDIAN_ACTOR`, `NEAR_MEDIAN_ACTOR` or `AT_OR_BELOW_MEDIAN_ACTOR`, adding `NOT_COMPARABLE` when the annotator item sets have no intersection and the median therefore mixes denominators. On our four tasks that chain returns `ABOVE_MEDIAN_ACTOR` on every one, because on all four the leader sits above the median annotator by the margins reported in §5.4; that chain is calibrated against known truth in §5.5, where it

returns the same token on synthetic pools built so that the raters have identical true accuracy, and where the identity of the leader depends on how the reference resolves its ambiguous labels — on one of the two references we build the leader is a rater that answers positive on every item while the real judges sit below it, and on the other the same judges lead by a wide margin. The words name the comparison rather than a readability verdict, so a released report is not quotable as if the rule had fired: the branch is identifiable from its token alone, and the `floor_basis` field (`median_of_actors` here, `judge_gold_leave_one_out` on the first chain) says the same thing in machine-readable form. This is a repair rather than a disclosure: both chains grew from one script, and the earlier shared vocabulary let a reader mistake a baseline comparison for a verdict.

4.3 Three rulers, and which question each answers

The three rulers answer different questions, and their outputs are not interchangeable. This distinction is the main reason a reported number is often wrong, not merely uncertain. Table 1 states the asymmetry

Ruler	Gold source	What the floor is	What a verdict means
Leave-one-out	judge consensus (2+ models)	accuracy of a gold generator with its own votes removed, the *maximum* over generators, since a leader must clear every generator rather than the average of them	whether the top of this leaderboard can be read at all
External human gold	human annotation	*undefined*; we report the median annotator instead	whether the leader beats a typical annotator
Latent variable (Dawid–Skene)	raw label matrix (no consensus label)	the agreement of the *weakest* annotator with the truth the model fits, i.e. the minimum over annotators — the opposite order statistic to (1), which is a maximum over generators — under a model that does not assume majority vote is truth	nothing: it feeds no verdict rule, only the signed comparison of §5.10

Table 1: The three rulers. The second row is the important one: under human gold the leave-one-out construction has no referent, because no model contributed to the labels. We return `NULL` for the floor and state the reason instead of substituting a number.

explicitly. The substitution is marked in every report as `floor_basis`, so that a reader who compares a leave-one-out margin with an annotator-median margin can see that the two are different measurements rather than two estimates of one.

4.4 Declining to estimate what the data does not pin

A latent-variable model of annotations (10) estimates a latent truth together with a confusion matrix per annotator. It requires enough annotators to identify those quantities. With two annotators and a single label per item, the latent truth and the two accuracies are three unknowns against two equations: every confusion matrix fits equally well, and the estimator returns a value determined by its initialisation rather than by the data. We check this before estimating and refuse to run when the annotation count is below three, recording the refusal in machine-readable output. This is the one place in the tool where a refusal is the correct answer, and we treat a silent fallback to an identifiable-but-wrong estimator as worse than no answer: an unreadable leaderboard is detectable downstream, an unjustified floor is not. The word *floor* deserves one fix here, because the two rulers take opposite ends of a distribution. Equation (1) is a *maximum* over gold generators, since a leader must clear every generator rather than the average of them; the latent arm of §5.10 is a *minimum* over annotators, the reproduction rate of the least reliable one. A minimum over annotators is not the same quantity read at the other end; it is a different quantity, so Table 8 names its two right-hand columns by what they compute — *agreement* — and §5.10 declines to accept any of the four values as a floor. Only the first of the two enters the decision rule (3); the second feeds no verdict, and we state that in Table 1 rather than leaving it to be inferred.

Before you cite a judge-gold leaderboard: five checks, in order. First, name the gold generator set and its size m; below three generators the leave-one-out floor is an agreement rate, and no margin over it is interpretable. Second, name the aggregation and the tie rule, and say whether they were chosen before the ranking was computed. Third, check that the floor and the leaderboard are computed on the *same* item set, not merely on sets of the same size. Fourth, disclose whether any ranked system contributed to the gold; if so the floor is optimistic by an amount this instrument does not estimate. Fifth, report the margin with its interval and the resolution the sample supports, and label the output with one of `READABLE`, `MARGINAL`, `UNREADABLE`, or the named refusal — never with a widened estimate. Each check is enforced by a gate in `audit.py` rather than left to the reader's care. The five are the operational form of the invariants of §2, restated here so the box can be read on its own.

Work published while this paper was in review measures two noise sources adjacent to ours. (2) re-runs pinned judges on standard chat benchmarks and reports large per-item flip rates on the close-call items that decide margins, so a judge's own verdict is not reproducible at temperature zero even at a fixed reported model version; that floor sits on the judge's verdicts, whereas the floor defined here sits on the reproduction of a gold definition, and the two are not one number. (15) measures error dependence across a bank of ten judges and finds that the effective number of independent votes is far below the headcount, and (19) reports the same compression for a nine-judge panel on natural-language-inference items, with the panel failing to beat its best single judge by a margin outside noise. (18) shows that a judge agrees with itself more than with other judges when configuration changes. These results bound what a panel can buy. They do not answer the question this paper asks, whether the gold a leaderboard is scored against can be reproduced by one of its own producers, and they leave the refused case we report untouched.

4.5 What these rulers do not claim

A floor measured this way is a statement about a ruler, not about the market it is holding, and the difference matters enough that we state the boundary before it is read across. Three families of claim sit outside what we measured, and we list them so that the silence is stated rather than left to be inferred.

A refusal is not evidence of indistinguishability. The clearest case is a perfect gold. If every generator labels every item correctly, the leave-one-out floor is 100%, and the rule refuses every board — including one on which a 90% system is plainly ahead of a 50% one. The refusal is correct about its own quantity and silent about the ranking: no leader can be shown to exceed a perfect producer. A board can therefore be informative and still fail this test, and a refusal must not be read as a claim that the systems are tied. What the rule certifies is the narrower statement that the leader clears the pool that made the gold; whether

the ordering below that level is resolvable is a separate question, assessed where the data support a paired interval and not established by the gate's refusal.

We do not measure Goodhart's law. When a measure becomes a target, the measure tends to stop measuring what it was built for (12); in the machine-learning setting this has been given a structural form in which optimising against a proxy introduces a bias whose size depends on how much the proxy is manipulated (14), and in social measurement the same distortion has been observed since (6). What we measure is prior to all of that: no contestant in our measurement was optimised against any gold definition we score it on. Our quantity is the noise floor of the instrument, the size of the disagreement a leader must clear before a difference can be attributed to the systems rather than to the labels, and it says nothing about how a difference that is clear of that floor moves under optimisation pressure. A leaderboard can be readable by this ruler and still be a measure that has stopped measuring, and vice versa; the two questions have different answers and we answer only the first.

We do not measure pairwise preference. Our corpora score a response against a gold label under (1), so they are checkable-answer tasks by the definition of the ruler. Pairwise preference — an Arena-style leaderboard in which a human chooses between two responses and scores are fitted to a Bradley–Terry model (8, 4) — is a different measurement with a different estimator, and no number in this paper is computed on it. The boundary is worth naming because the two share a noise-floor problem: a pairwise judge that is itself a model has its own reproducibility limit, and a gap below that limit is not a ranking. Whether that limit binds on any particular preference board is a measurement we do not make. We note only that the platform's authors already report stability analysis of their own, which is the same question asked of a different instrument.

We do not measure post-training. The reliance of alignment training on model-generated preference judgements (26, 28) means a floor on those judgements passes through to the objective being optimised. Our measurement does not touch that chain: we score contestants after the fact, on labels produced by two generators, and we do not train anything. The consequence we do claim is narrower and is about reading, not about training — a difference smaller than the resolution of the corpus is a number rather than a measurement, and that holds for whatever the number is used for afterwards.

What the rulers do claim is a protocol, and it is checkable rather than argued: name the aggregation, name the convention that resolves a disagreement, disclose the actor overlap between the gold generators and the contestants, give the floor's item set and its size, and report the resolution the sample size supports. The five invariants of §2 are that protocol in executable form, and the reviewer's checklist is only the question of whether a published number comes with them.

4.6 What the rulers ask of a benchmark

The preconditions are the five invariants of §2, stated there once with the gate or the reported field that enforces each, and we do not restate them here. What is worth adding is their status: they are not a proposal for a different benchmark. They are the conditions under which the numbers a benchmark already publishes can be read as measurements, and each is a gate in Algorithm 1 or a field in the released reports, so an author can check them against a run rather than against our prose. Two of the five stop a verdict, two are disclosed alongside one, and one is a declared parameter — the same split this section's refusals are built on.

5 Measurements

We measure on three public corpora, and the count matters because they support different rulers. The first two are two exports of one set of four public snapshots, retrieved on 2026-10-01:

```
typesafeAI/evalsafe-invoice-processing
typesafeAI/evalsafe-customer-service
typesafeAI/evalsafe-security-incidents
typesafeAI/evalsafe-agent-trace-observability
```

The releases' default reference is a consensus of the two contributor models, their probabilities averaged with equal weight; the two contributor columns are what this paper treats as the two gold generators, and no human labels enter either export. The repository's commit at retrieval was not recorded; the twelve parquet files we used are pinned by SHA-256 in a digest record registered in the manifest, and the vote tables this paper measures on are pinned in the same manifest. The decision-level export contains 1,524 policies and nine contestant systems. The question-level export contains 14,292 questions from the same four sources, with the same nine systems and per-answer probabilities. The third is a set of four annotation tasks with human gold (`Rohithreddybc/judgesense-benchmark`), in which 25 judge models each label 500 to 1,000 items; that set is documented by its own authors as measuring whether a judge reproduces its verdict under rephrasing rather than whether the verdict is correct, so using it here goes beyond the use it advertises (§8). The first two are where a model-generated floor is available as published; the third is not, because its gold is human, which is why §5.6 has to impose a model-generated gold on it rather than find one there, and why the verdicts of that section carry the caveat they do. Table 2 collects what each ruler returns on each of these substrates, and its caption names the rows rather than numbering them, because two of the five rows are one structural fact observed on two different sources of votes.

Corpus	Gen.	Majority-vote floor	Latent-variable arm
decision (1,524)	2	`DOMAIN_MISMATCH` (refusal)	not identified at $m = 2$
question (14,292)	2	`DOMAIN_MISMATCH` (refusal)	not identified at $m = 2$
human gold, as published	none	no model-generated gold	refused by identifiability screens
human gold, 25-model gold	25	3× `UNREADABLE`, 1× `DOMAIN_MISMATCH`	four values printed, none accepted as a floor (§5.10)
public judge panel, votes released	17	`UNREADABLE` at margin 0.00	not fitted on this panel

Table 2: What each ruler returns on each substrate, so that a reader can see at a glance which rows are refusals and which are verdicts. Rows are named here rather than by position, because the table has five. The *25-model gold* row is the construction of §5.6: its verdicts are `UNREADABLE` on factuality, coherence and relevance, each at a margin of exactly 0.00 points, and a `DOMAIN_MISMATCH` refusal on preference — refused by the domain gate rather than by its floor, whose margin there is also 0.0 — and therefore not drawn in Figure 1. The *public judge panel* row is §5.7, the one reading whose votes were published by somebody else and the one we report as not reproducible offline, because we cite those votes rather than redistribute them. Those two rows are alike in the way that matters most: in both, the ranked field *is* the generating pool, so a leave-one-out floor there is a self-reproduction rate rather than an agreement rate, and Proposition 4 makes its equality to the leader exact rather than approximate. The two rows above them have two generators against nine contestants, so their pool and field do differ, and what stops them is a domain mismatch rather than a collapsed floor. The *human gold* rows share one annotation set, and their two latent-variable cells state one negative result about it: §5.10 fits the estimator to that set, prints the four agreement values of Table 8, and accepts none of them as a floor in the sense of (1). The *public judge panel* row was not fitted with that estimator.

5.1 The decision corpus yields no verdict under either tie rule

The decision-level corpus is where we expected a clean verdict, and it is where the measurement refuses to produce one. Its structure is small enough to state exactly: 1,524 items, of which 1,520 carry a vote from each of the two gold generators; the remaining four carry a single vote, so the leave-one-out estimate is undefined on them and they leave the floor's domain for that reason rather than through any label being missing. On 436 of the 1,520 paired items the two generators disagree. An earlier version of this section reported a leave-one-out floor of 70.23%, a leader at 73.19% and a margin of +2.96 points with a verdict of READABLE. Those numbers are withdrawn. They came from a tie rule that silently resolved 436 of 1,520 items by sorting the two answers as strings, which is an arbitrary rule that also moved the leader; we no longer report an outcome under it.

What the corpus supports is the following. Under the default rule, which treats a disagreement as unmeasured, the leave-one-out floor is defined on all 1,520 items for which both generators vote, because the

leave-one-out estimate of the gold definition is undefined only where removing a generator leaves no vote behind. The leaderboard, however, is scored only on items that carry a gold label, and on this corpus 436 of those items do not, because a disagreement produces no label under this rule. The floor therefore lives on 1,520 items and the leaderboard on 1,084, and a margin between them is a difference of two different denominators. The tool reports UNDECIDABLE_DOMAIN_MISMATCH and names the mismatch rather than producing a number. The two measurements are still available and we state them separately: the leader scores 86.48% on the 1,084 items it is scored on, and the floor is 70.23% computed on the 1,520 items that survive a generator's removal, as an unweighted mean over the corpus's four slices rather than as a pooled rate (a pooled rate would be 71.32%; §5.3 gives both). That 86.48% is not an estimate of the leader's accuracy on the corpus: the 1,084 items are exactly those on which the two generators agree, and the same system scores 73.19% on all 1,520 under the forced rule. The 13.3-point difference is the selection effect of conditioning on label availability — it implies roughly 40% accuracy on the 436 disagreement items — and every score in this paper computed on a labelled subset carries it. We do not read a verdict off these two numbers: the released report carries their difference and its interval as reference, which is what a refusal does here (§4.1), but the terms sit on item sets of different size, and a threshold is not applied to a difference of that kind. A paired bootstrap of their difference would resample an item set that the two terms do not share, so the resampled quantity is not an estimator of anything in the corpus; printing an interval around it would give a refused comparison the appearance of a measurement, which is the one thing a refusal exists to prevent; where a refused row does show an interval, it is printed as reference and is never the basis of the verdict.

The refusal is the finding here, and the two item counts say why the comparison the rule would need is not available on this corpus as scored. We state one aligned variant rather than leave a reader to supply it, because it is the natural objection to a domain refusal: restrict both terms to the $1,084$ items on which the two generators agree, which satisfies the membership test without invoking any tie rule. That variant aligns the denominators and costs the quantity its content. An item of the intersection carries the same answer from both generators by the very condition that placed it there, so removing either generator leaves the other's vote, which is the answer being scored; the reduced gold therefore agrees with each generator on every item of that set, the maximum over generators is a perfect score, and on that domain every system on the board sits at or below its own floor, so the rule returns no readable verdict for any of them at any non-negative margin.

We report that as a consequence of (1) under two generators, in the same family as Proposition 1 and not as a reading, and it is the reason the refusal of this section is not an artefact of the tie rule we declared. The two ways of making the denominators agree each take something away: the intersection collapses the floor to a tautology, the forced rule hands the label to the string order of an answer.

A reader who wants the same arithmetic for the paper's other readings has it in Table 3, which sets the item count, the items that carry no label under that row's tie rule, the floor's domain, the items the leader is scored on, and the membership test of Algorithm 1 side by side for each row.

Forcing the two denominators to agree requires resolving every disagreement, and the only rule available that does so is the arbitrary one: it sorts the two answers as strings. Under it every item carries a label, the leaderboard and the floor both live on 1,520 items, and the denominators finally match. What that alignment costs is the quantity itself. With two generators the leave-one-out estimate of the gold definition is, after removing a generator, the surviving generator's own answer, and a generator always agrees with itself, so the leave-one-out value is an inter-generator agreement rate, identical for both generators by construction, and taking the maximum over generators is a no-op. The tool reports UNDECIDABLE_FLOOR_DEGENERATE and declines to convert a high agreement rate between two generators into a claim that the leader is or is not readable. Under this rule the leader scores 73.19% on all 1,520 items against the same floor of 70.23%, a margin of +2.96 points with interval [−0.14, +6.27], which contains zero; we record the number because it is what the tool printed, and we decline to read it as a verdict, because the two terms measure different things and because the interval does not exclude zero.

Two negative results, then, from the corpus we had most confidence in. The first is that the tie rule is not a formatting detail: on 28.6% of the candidate items it decides whether a verdict exists at all. It does not

Reading	items	no gold	floor set	leader set	same?	arm returns
Decision board, ties unmeasured	1,524	436	1,520	1,084	no	`UNDECIDABLE_DOMAIN_MISMATCH`
Decision board, ties lexicographic	1,524	436	1,520	1,520	yes	`UNDECIDABLE_FLOOR_DEGENERATE`
Question board, majority gold	14,292	799	12,406	12,330	no	`UNDECIDABLE_DOMAIN_MISMATCH`
Question board, mean-prob gold	14,292	799	12,406	12,330	no	`UNDECIDABLE_DOMAIN_MISMATCH`
Public judge panel	5,918	0	5,376	5,376	yes	`UNREADABLE`
25-model gold, factuality	500	2	486	486	yes	`UNREADABLE`
25-model gold, coherence	500	13	445	445	yes	`UNREADABLE`
25-model gold, relevance	1,000	4	987	987	yes	`UNREADABLE`
25-model gold, preference	520	0	511	511	no	`UNDECIDABLE_DOMAIN_MISMATCH`

Table 3: Where each reading's numbers are counted. Each cell is read from the report field of the same name: `n_items` for the item count, `gold_tie_items` for the items on which that row's tie rule produces no label, `loo_domain_size` for the domain of the leave-one-out floor, `leaderboard_scored_items` for the items the reported leader is actually scored on, `same_domain_as_floor` for the membership test, and `verdict` for the word the arm returns. *Same?* is that membership test rather than an equality of counts, because two sets of the same size can carry different members: the preference row has 511 items on each side and different members, and the arm refuses it on that basis. The table prints counts; the membership test behind the last column uses the digests stored in the reports rather than these counts. The panel arm's report carries no candidate-item field, so its row is read off the four fields it does carry.

decide who leads on this corpus: `claude-opus-5` is first under both rules. What the rule does decide is what that first place is worth, 86.48% against 73.19%, and it exchanges third with fourth and sixth with seventh. The corpus where the rule changes the identity of the leader is the question-level one of §5.2. The second is that the threshold $\delta = 2.0$ of (3) is unreachable in practice on this corpus, since neither rule produces a verdict at any threshold; under the forced rule the margin sits +0.96 above the threshold while its interval spans both signs, so a verdict at $\delta = 2.0$ would rest on a difference from zero that the resampling does not resolve. We therefore do not report a δ-sensitivity analysis here; an earlier version claimed that at $\delta = 0.0$ the decision corpus would remain READABLE, which was both unverified and in tension with the interval that the same section reported. What would settle this corpus is a third gold generator: with three, removing one leaves two, the leave-one-out quantity stops coinciding with agreement, and the tie between the remaining two is decidable by construction.

5.2 The same data, two gold definitions

The question-level corpus carries two gold generators as well, and this turns out to matter more than the choice between gold definitions. Both definitions leave the measurement undecided, and for a reason that has nothing to do with which definition is used: on 13,104 items each carries a row from each of the two generators, so a disagreement has no strict majority to fall back on. One of those items is not a pair but a quartet: `customer-service::retention-charged-after-cancel/t0::intent_pair::intent_pair` carries two rows from each generator, which is why "exactly two annotations" is not literally true of every item and why a row count and an item count part company there. On 12,305 items the two agree and both definitions return that answer; on 799 they disagree, and with two generators a disagreement leaves no strict majority to fall back on. Under our default rule those 799 items leave the gold table, which takes the labelled set to $12,305$ items. The leave-one-out floor is not computed on that set but on the intersection of the two generators' own reduced domains, which the report prints as `loo_domain_size` $= 12,406$ items; the two per-generator entries in `floor_per_generator` are `loo_n` $= 12,903$ and $12,608$, and those are scoring *rows*, not item sets — they differ from the item counts of the two reduced domains by the duplicated item, so the intersection does not follow from them by subtraction and we do not compute it that way. The leaderboard is scored on the $11,586$ that carry a label inside it, and the tool reports

UNDECIDABLE_DOMAIN_MISMATCH for both definitions. Forcing the denominators to agree means resolving every disagreement, and the two definitions then resolve them by different but equally arbitrary rules: majority by sorting the answers as strings, mean probability by comparing the arithmetic mean of the two confidences. The gold definitions then differ, and so do the leaderboards built on them, because each definition also decides which items carry a label at all. Under the forced rule every two-annotation item is labelled, taking the labelled set to 13,104 and the leaderboard's own scored set to 12,330, while the floor stays on the same 12,406-item leave-one-out intersection, so the two terms do not end up on one denominator after all: the instrument returns UNDECIDABLE_DOMAIN_MISMATCH under both definitions. Table 4 reports the majority leader at 87.91% against a leave-one-out floor of 89.48% (slice-equal; the pooled rate is 93.56%), a margin of −1.57 points with a 95% interval of [−2.36, −0.79] that excludes zero. The instrument prints those readings and declines to decide on them. An earlier version of this sentence recorded the corpus as UNREADABLE under this rule, which is the error this paper exists to describe: a refused comparison is not the negative end of the scale, and a leader sitting below a floor measured on another item set has not been shown to sit below anything. The withdrawal is registered in Appendix A. Under mean probability the leader is `gpt-5.6-sol` at 89.01% on 12,330 items against the same floor, a margin of −0.47 points with interval [−1.29, +0.35], which contains zero; we report it as a refused comparison for the same reason as the majority one, and not as a measurement of judge accuracy, for the reason given below. Neither difference becomes a verdict, because the forced rule is arbitrary and because the two disagree about almost everything above it.

The number of moved ranks is itself a reading of a report rather than a property of the corpus: `n_rank_flips` is 4 under the rule that leaves a disagreement unmeasured and 3 under the forced one. The two gold conventions disagree about the leader outright in the default report, where `claude-opus-5` leads the majority board and `gpt-5.6-sol` the mean-probability one, and they agree on the leader in the forced report, where `gpt-5.6-sol` leads under both. Which system holds which place is itself a reading of one carrier, so we name the carrier for the positional and calibration sentences that follow: they are read from `evidence/audit_both.json`, under the rule that leaves a generator disagreement unmeasured, whose majority board puts `claude-opus-5` first, `gpt-5.6-sol` second and `jev-1.13.0` eighth. Under the forced rule that board exchanges the top two and exchanges fifth with sixth, so `gpt-5.6-sol` leads at the 87.91% of Table 4; `jev-1.13.0` is eighth on every one of those four boards; an earlier version of this sentence had it rising to first once the confidences were averaged, which was a reading of a defect rather than of the corpus, and it is withdrawn in Appendix A.

The mean-probability leaderboard is not a measurement of judge accuracy, and reporting it as one would be the worst error available here. Both generators put their confidence on the answer they themselves voted for, so the sign of the difference between the two confidences decides every disagreement; that is a difference-of-signs rule, not a confidence-weighted vote, and on the undecidable items it selects by a gap that is far from negligible: over the 709 of the 799 disagreement items that the resolution instrument reports as measured (`evidence/tie_rule_resolution.json`; the 90 outside its input are named as a gap below and in §8) the median difference between the two confidences is 0.250, only 2.82% are decided by a gap below 0.01, and the larger confidence belongs to the same generator on 92.8% of them. That carrier also reports 0 disagreements skipped for a missing probability, so the 709 it measures is not the 799 the gold leaves unlabelled, and the carrier we ship does not say which 90 items are outside its input or why; we state that as a gap in our own instrument rather than close it with a guess, and §8 counts it. The rule is therefore a vendor weighting rather than a tie broken at the third decimal. Its leaderboard therefore reports what the arithmetic happens to produce rather than what either generator believes.

We checked whether the confidences themselves are at fault, and they are not: across the nine contestant systems on that board the expected calibration error against the majority gold ranges from 0.0084 to 0.0961, and mean confidence tracks accuracy closely enough that no system's scores are being generated by a generator certain of a wrong answer. Three denominators are live in this paragraph and they are not one object, so each is named where it first appears. The calibration block is computed on the default-rule carrier `evidence/audit_both.json` over the rows that carry a confidence rather than over the items a board is scored on (11,939 rows for `gpt-5.6-sol`), and its accuracy is a fraction on 0 to 1; that same board's macro accuracy for that system is 93.37 points, which is not the 87.91 of Table 4 because the table is the forced-rule

board; and the accuracies quoted below are the first of these three quantities, not a cell printed anywhere. The best-calibrated system, `gpt-5.6-sol`, reports ECE = 0.0084 against an accuracy of 0.9599; the worst, `claude-haiku-4-5`, is overconfident by 0.0961 points of expected calibration error, with a Brier score of 0.1279 against an accuracy of 0.8378 and a mean confidence of 0.9338 that exceeds what it gets right — the largest such excess of the nine. The majority leader on that board, `claude-opus-5`, goes the other way, with mean confidence 0.8905 below an accuracy of 0.9601 (ECE = 0.0701, Brier = 0.0357): it says "true" less often than it is right. So the confidences are usable, and the numbers a disagreement would have supplied are the ones the rule discards. The collapse therefore belongs to the rule that consumes the confidences, not to the confidences it consumes.

We give the numbers because the contrast between the two conventions is the point of this comparison, and we label the mean-probability leaderboard as not a measurement of accuracy wherever it appears. What survives both conventions is the ranking comparison: the board reorders under each of them, and neither convention supports a verdict. We therefore report the table as a sensitivity result, with the verdict column marked undecided in both rows.

System	majority	mean-prob	Δ	rank
`gpt-5.6-sol`	87.91	89.01	+1.10	1→1
`claude-opus-5`	87.76	87.74	−0.02	2→2
`gpt-5.6-terra`	85.85	86.82	+0.97	3→3
`deepseek-v4-flash-0731`	85.67	86.63	+0.96	4→4
`claude-sonnet-5`	85.25	85.47	+0.22	5→7
`deepseek-v4-pro-0813`	84.96	86.44	+1.48	6→5
`gpt-5.6-luna`	84.87	86.04	+1.17	7→6
`jev-1.13.0`	84.06	84.76	+0.70	8→8
`claude-haiku-4-5`	75.28	76.11	+0.83	9→9

Table 4: Macro accuracy under two gold definitions on the same 14,292 questions, each scored on the items that carry a label under the corresponding definition. This is the *forced*-rule board: majority ties are resolved by answer string order and mean-probability ties by the mean of the two confidences. Under the rule that leaves a disagreement unmeasured the majority board is led by `claude-opus-5` rather than `gpt-5.6-sol`, and 4 rather than 3 of the nine systems change rank. Δ is computed before rounding; three of nine systems change rank here while the leader holds first place under both definitions; both definitions return an undecided verdict because the two annotations per item leave the floor and the leaderboard on different item sets. The mean-probability rows are not a measurement of judge accuracy, for the reason given in the text.

5.3 Aggregation changes absolute scores

Macro accuracy averages over items; micro accuracy pools all rows and weights them by count. Because the question corpus is assembled from four annotated sources of unequal size, the two differ for every system, by between 2.70 and 5.02 points on the majority-gold leaderboard, while the leader-to-tail span changes by only 0.36 points (12.81 macro against 12.45 micro). Micro is the higher number throughout under majority gold, because the larger sources are also the easier ones; micro remains the higher number under mean-probability gold as well, where the gap widens to between 4.19 and 5.91 points, and the leader-to-tail span is 12.90 macro against 12.41 micro. The tie rule of §5.1 reaches this quantity too: on the same nine systems, under the rule that resolves a disagreement by string order rather than leaving it unmeasured, the largest macro-to-micro gap on the majority board is 6.75 points (`evidence/jev_question_lexicographic.json`) rather than the 5.02 above — so even the *size* of the aggregation effect is a reading of a convention, and §1's "up to 6.75 points" is the forced rule's number while §5.3's own 5.02 is the default rule's. This has a practical consequence for citation: two groups computing "the score" of the same model can report values differing by six points while both are correct, and the difference is a property of the aggregation rather than of the model. We therefore require a reported score to name its aggregation, and the tables in this paper report macro throughout. The same requirement reaches the floor itself, and a reader who divides the counts printed in §5.1 will not recover it: both floors are unweighted means over the four slices. `make_floor_aggregation.py` re-

derives them from the released vote tables and writes nothing unless its slice-equal value matches the shipped report's `ceiling_loo_macro` for every carrier it names; on the decision corpus that value is 70.2323% slice-equal against 71.3158% pooled, because 1,084 of the 1,520 paired items carry the same answer from both generators, so the 1.08-point gap is the weighting and not the data. On the question corpus it is 89.4819% slice-equal against 93.5596% pooled: of the 12,406 items on which both generators return a non-empty answer, 11,607 carry the same one, a gap of 4.08 points. Both readings, the per-slice counts they come from, and the carrier-by-carrier cross-check are in `evidence/floor_aggregation.json`, and the instrument's own battery contains a case in which a shipped ceiling is replaced by the pooled value of the same corpus, which it turns red.

A paired test across the nine systems says the direction of that gap is systematic rather than anecdotal: the mean macro-to-micro difference is -3.58 points ($t(8) = -14.90$, $p < 0.001$) under majority gold and -4.89 points under mean-probability gold, -1.97 points on the decision corpus under the default rule ($p = 0.015$) and -0.95 under the forced rule ($p = 0.136$), the only one of the four that does not separate from zero; on the m25 construction, the human-gold chain and the panel the two aggregations are the same number, because with a single slice the equal-weight mean and the pooled mean are one quantity, and the paired test degenerates there rather than reporting a null result (`evidence/strict_review_followup/EC39_macro_micro_paired_test.json`).

5.4 With human gold, the floor is undefined rather than different

On the 25-judge corpus we have human annotation, so the leave-one-out construction of (1) has no referent. We report the median annotator instead and mark the substitution. That median is a mean over annotators and so is only a median on a common denominator when every annotator answered the same items, which none of these tasks guarantees; we therefore compute it on the intersection of annotator item sets and report the intersection size alongside it. On all four tasks the leader sits above that median: by 13.60 points (500 common items), 9.20 points (500), 4.50 points (1,000) and 12.50 points (64 of 520). We report those four differences and we do *not* report a readability verdict from them, because the reference point they are measured against is not a noise floor. The released reports carry the token this chain emits in a field named `verdict`. We now regard that name as wrong: the field is a point comparison with no interval, its null rate is measured in §5.5 and §8, and no verdict should be read from it. The next release of the reports renames the field `point_margin_token` and carries the per-task permutation rate beside it.

The reason is structural, and it is the same structural fact that the self-grading caution in this section records. The median annotator on these tasks is the median of the same 25 actor models that the leaderboard ranks, and the leader is the maximum of those same 25. "The leader clears the median" is therefore a statement about the spread of one set of 25 numbers, and it is true of any leaderboard whose field is not degenerate. It is not evidence that the leader's labels are reproducible, because the quantity on the right-hand side was computed from the field on the left. A readability verdict needs a floor whose value does not come from the population being ranked, and on this corpus no such quantity exists: the labels are human, so no model contributed to the gold and the leave-one-out construction has no referent. We could substitute the median, but substituting it changes the question—from "does the leader clear the label producers" to "is the leader above the middle of its own field"—and the second question is answerable without any floor at all.

The direction of the resulting bias is worth stating rather than leaving implicit, because it runs against the reading. If self-grading inflates a model's apparent accuracy against human gold, it inflates the maximum of the 25 more than the median, so the margin we report is if anything an overstatement of the distance between the leader and a floor that a non-overlapping annotator set would set. We therefore report these four margins as an upper bound on any floor-based reading, and not as a result about readability.

One correction belongs here for the record. An earlier version reported the fourth margin as 5.77 points; that figure was computed before the four margins were placed on a common item set, and on the preference task the alignment moves it to 12.50 because one annotator who answered a different question set had been deciding a third of the value. The corrected number is not a better estimate of the same quantity—it is a different and better-defined one. Four cautions attach to this table. The first two bear on how the table

should be read; the last two bear on how its counts and intervals should be quoted, because that table is where a reader is most likely to take one number for another.

The judges are the contestants. On all four tasks the set of models appearing as annotators is identical to the set appearing as contestants: the intersection is 25 of 25, and on every item the annotation recorded under the two roles is identical. On all four tasks every one of the 25 judges is grading work it produced itself, so the floor is constructed in part by the systems it is meant to bound. We detect this in the tool by taking the intersection of the actor sets across the system and annotator files, and we report the overlap in the machine-readable record for each run; a reader who wants an unbiased floor needs annotators that are not in the field being ranked. We report the numbers anyway because the alternative—silently reporting a floor that the participants helped build—is worse, and because the next three cautions bound how much of the table survives the problem.

$\delta = 2.0$ **is below this measurement's resolution.** The median minimum detectable difference is 4.55 points on factuality, 7.40 on coherence, 3.08 on relevance and 15.63 on preference, every one larger than the declared threshold of (3), and the approximate power at $\delta = 2.0$ is 0.14, 0.08, 0.25 and 0.06 in the same order. Neither number comes from a closed form. For each system the report stores the paired case-level bootstrap interval $[\ell, u]$ of its difference from the leader, and `mde_from_ci` returns its half-width $(u - \ell)/2$, which is that difference's standard error scaled by the two-sided critical value at $\alpha = 0.05$; the median over the systems on that task is the figure printed above, and every one of those four figures is a resolution statement about the 24 system-versus-leader contrasts on that task rather than about the leader-versus-median-annotator margins of §5.4, for which this release computes no paired interval (§8), and `power_at_delta` recovers the standard error as the half-width divided by the two-sided 95% normal critical value and inverts it at that same critical value, so the two printed quantities are one estimator read twice and the convention is stated in words rather than left to the reader: a one-sided inversion would print a smaller power for the same four intervals, which is a different quantity, not a different measurement. Clustering enters through the resampling unit rather than a divisor: cases are drawn whole, so the deff $= 1 + (k - 1)$ICC correction named two paragraphs above is not applied to these figures, and the design effect that the reports also carry (`clustering.design_effect_judges`) is a measurement of how much clustering there is, not a term in the MDE.

What the printed MDE therefore means is the smallest difference this design would have separated at $\alpha = 0.05$, not the *a* priori power-based minimum detectable effect: the latter replaces the single critical value $z_{\alpha/2}$ by $z_{\alpha/2} + z_\beta$ and is larger by the factor $1 + z_\beta / z_{\alpha/2}$, so every MDE in this paper is the optimistic end of that pair. At this power the δ-tier verdict carries no statistical force on any of the four tasks: the four ABOVE_MEDIAN_ACTOR readings printed above rest on margins far above δ, which is why the resolution problem does not void them of itself, but a corpus whose leader-margin happened to land near 2.0 would be declared undecidable by us, not MARGINAL. A "not significant" result here means the measurement does not resolve the difference, not that no difference exists.

Three counts answer three questions about the same 24 **comparisons, and one family is corrected.** Each task carries 24 paired comparisons against the leader. Read one at a time at $\alpha = 0.05$, the p-values give 23, 21, 20 and 8 significant comparisons on the four tasks; under Holm–Bonferroni, which controls the family-wise error rate without an independence assumption, the corrected family rejects 23, 14, 18 and 3. The second list is a subset of the first: Holm can only tighten the criterion as it steps up the ordered p-values, and the tool checks that containment instead of assuming it. The 70.8% belongs to neither list: it is $1 - (1 - \alpha)^{24}$ at $\alpha = 0.05$, the probability that the uncorrected procedure rejects *at least one* null when *all* 24 nulls are true. That is a property of the procedure under the global null, not the proportion of our rejections that are false, which would require knowing how many of the nulls are false and is not computable from these data. A third count comes from the intervals the table prints. Those are per-comparison, case-clustered paired bootstraps and are not adjusted at all, so they belong to no corrected family; the count they give is the next bullet's, and it is a third number answering a third question. On the preference task the three readings are 8 unadjusted p-values, 3 corrected ones and 16 intervals that cover zero, and what they show is that the questions differ, not that one of the three is the answer. We correct the within-task family and no other: the corpora of this paper are not a family, because each is compared against its own floor, so there is no shared null across them to control.

The leader-to-annotator margin is not a floor, and the number of systems indistinguishable from the leader is substantial. The count is 1 of 24, 6 of 24, 7 of 24 and 16 of 24, the last on 64 common items, where most of the field lies inside the interval. Reporting "the best model scores 88.00" without that count overstates the resolution of the measurement.

The leader-versus-median contrast has its own interval. The release recomputes that contrast with the leader held at its full-data identity and the median recomputed inside every draw, on 2,000 case-clustered draws (`evidence/ds_median_interval.json`). The margin is +13.60 points on factuality (interval $[+10.40, +17.20]$), +9.20 ($[+7.00, +13.80]$) on coherence, +4.50 ($[+1.30, +8.00]$) on relevance and +12.50 ($[-4.69, +31.25]$) on preference. Three of the four intervals exclude zero; the preference interval does not, because that task has 16 cases, so we report its margin as unresolved rather than as a spread that clears zero.

Coverage at that cluster count is itself measured rather than assumed, and it is not free: re-running the four tasks at $n = 16$ cases, the smallest resampling shape these corpora present, the paired interval covers the full-data margin in 0.900, 0.960 and 0.892 of five hundred resamples on factuality, coherence and relevance against a nominal 0.95, preference's intervals reach 0.998 only by averaging 24.16 points wide, and the above-median reading on relevance holds in 0.68 of the resamples — at sixteen units the interval is part of what fails, not a remedy (`evidence/strict_review_followup/EC37_bootstrap_coverage.json`).

Two further quantities belong with this table because they determine how its intervals should be read. The items are clustered: an annotator answers several items drawn from the same case, and on these corpora a case contributes two items on factuality and coherence and four on relevance and preference, so the observations are not independent. The agreement coefficient we report for this purpose is the intraclass correlation in its two-way random, absolute-agreement, single-measurement form, $\mathrm{ICC}(A, 1)$, computed over the items and judges that each task actually shares. It is a coefficient of substitutability rather than of consistency, because the question is whether one judge's score can stand in for another's, not whether the two order items the same way. We report the point estimate with an item-level bootstrap interval, and we name the type explicitly because the conventions are not uniform: this is an *inter-judge* coefficient over binary hit measurements with 25 judges per task, not a test–retest reliability, because these corpora contain no repeated measurement of the same judge on the same item. That distinction governs how the number should be read. The relation to this paper's own object is stated once, because a reliability vocabulary invites the wrong reading: the coefficient is not an input to any floor here. The leave-one-out floor of (1) is a generator's score against a reduced gold table, computed without it, and neither Algorithm 1 nor (3) reads an ICC term; the auditor's code path makes that checkable, since the coefficient is written only into the `clustering` block of the report while the decision reads the floor, the leaderboard and their difference. What the coefficient governs is how wide a margin a design of this kind could resolve, which is why it appears beside the table rather than inside the rule.

The coefficient takes the values 0.30 (interval $[0.27, 0.34]$) on factuality, 0.19 ($[0.17, 0.22]$) on coherence, 0.46 ($[0.42, 0.50]$) on relevance and 0.42 ($[0.32, 0.51]$) on preference. Every one of those intervals excludes zero, so the judges are correlated; but none is near the value one would need in order to call the measurements interchangeable, and item heterogeneity inflates these estimates, so we treat them as an upper bound on agreement rather than as evidence of reliability. We give the estimator rather than only its output, because a coefficient whose decomposition is hidden is not auditable: with M the $n \times k$ table of item scores, r_i the mean score of item i and c_j the mean of judge j over a grand mean $\bar{x}$, $\mathrm{MS}_{\mathrm{row}} = \frac{k}{n-1}\sum_i (r_i - \bar{x})^2$, $\mathrm{MS}_{\mathrm{col}} = \frac{n}{k-1}\sum_j (c_j - \bar{x})^2$, and $\mathrm{MS}_{\mathrm{err}} = \frac{1}{(n-1)(k-1)}\left[\sum_{i,j}(M_{ij} - \bar{x})^2 - k\sum_i (r_i - \bar{x})^2 - n\sum_j (c_j - \bar{x})^2\right]$, so that

$$\mathrm{ICC}(A, 1) \;=\; \frac{\mathrm{MS}_{\mathrm{row}} - \mathrm{MS}_{\mathrm{err}}}{\mathrm{MS}_{\mathrm{row}} + (k-1)\,\mathrm{MS}_{\mathrm{err}} + k\,(\mathrm{MS}_{\mathrm{col}} - \mathrm{MS}_{\mathrm{err}})/n},$$

which is the form implemented in `icc_two_way_absolute_agree` (`audit.py`) and is what makes the word "absolute" mean that a judge who is offset by a constant is penalised rather than excused. Two numerical details are reported because a coefficient that hides them is not checkable. The denominator can in principle be made up of subtracted terms, so the error mean square is negative and the estimate is not inside $[0, 1]$; that did not happen on these four tables, where $\mathrm{MS}_{\mathrm{err}}$ is 0.0969, 0.1835, 0.0736 and 0.1388, and the point estimates are all interior.

The code therefore applies no truncation to zero and no clipping to $[0, 1]$, and reports the raw decomposition in the machine-readable record, so that a negative value would surface rather than be rounded into a plausible one. The interval is item-level: the resampling unit is the item, with 2,000 bootstrap rounds drawn from a fixed-seed generator (`np.random.default_rng`, seed 20261001, 64-bit PCG64), and a round that leaves fewer than 50 finite estimates does not contribute; if fewer than half the rounds are usable the routine reports insufficient rounds rather than a padded interval, because a confidence interval assembled from degenerate resamples is worse than an absent one.

On these four tasks all 2,000 rounds were usable. The latent-variable fit reported below runs to its own defaults, which stop on the largest change in any confusion-matrix entry falling below 10^{-6} after at most 200 iterations; it converged at iteration 20, 32, 16 and 10 on the four tasks. The tighter 10^{-10} and 500 settings appear only in the synthetic sweep of §5.12, which is a different construction and is labelled as such. The design effect each implies, deff $= 1 + (k - 1) \cdot \mathrm{ICC}$ with k the number of judges, runs from 5.67 to 12.09, taking the values 8.31 (factuality), 5.67 (coherence), 12.09 (relevance) and 11.06 (preference).

The design effect is a statement about how many independent units there are, so we give the effective sample size directly rather than letting the reader derive it: the case is the resampling unit, and the number of cases is 250 of 500 items on factuality, 250 of 500 on coherence, 250 of 1,000 on relevance and 16 of 64 on preference. The last of these is the one that matters for reading the table: sixteen independent units is what limits the resolution of that task, not the sixty-four items. Both bootstrap units are reported for that reason. The two units resample the same contrast, the spread between the best and the weakest annotator on the common set (not the four margins below), and they agree in sign on all four tasks; the case-level interval is the wider of the two every time, and each pair below is printed item-level interval first and case-level interval second so that the claim can be checked by subtraction: $[24.80, 33.40]$ against $[23.60, 34.60]$ on factuality, $[11.00, 22.60]$ against $[9.20, 24.40]$ on coherence, $[34.80, 41.50]$ against $[32.60, 43.20]$ on relevance and $[14.06, 50.00]$ against $[14.06, 51.56]$ on preference. We report both because choosing one unit is a choice the reader should make knowingly.

The median is a single number and the reader should not have to take it on trust, so we also report the distribution it comes from, as minimum, second-smallest, median and maximum over the annotators on the common item set. On two tasks that distribution is unimodal and the median sits in the middle of the field: 67.00, 67.80, 82.40 and 96.00 on factuality, and 28.60, 28.60, 36.20 and 45.40 on coherence. On relevance it is not: 49.80, 77.70, 83.50 and 88.00, where the smallest value is 27.90 points below the second-smallest, so that end of the field is one annotator rather than a distribution.

On preference the distribution is continuous but the median is not the centre of it: 39.06, 42.19, 59.38 and 71.88, with five annotators tied at the median, a gap of only 3.12 points between the smallest and the second-smallest, and a median that sits 17.19 points above that pair and 12.50 points below the leader. Both tails are explained by the same mechanism, and it is not grading ability. The three-valued label space these tasks use includes a "cannot tell" option, and an annotator that selects it has not answered the question; we therefore treat such an item as *missing* rather than as an error, removing it from both the numerator and the denominator of that annotator's accuracy. The choice is not cosmetic, because it moves the median. Counting abstentions as errors would push every annotator's score down and would push one annotator's score down much further than the rest, which moves the median as well as the tail; counting them as correct would do the opposite. Reporting the rate is what lets a reader undo either choice. On relevance the smallest value belongs to an annotator whose answers fall outside the gold label set on 41.7% of the items, against 9.2% for the next lowest; on preference the same measurement gives 26.6% against 17.2%. Two of the four tasks therefore have a floor whose lower edge is set by how often one annotator declines to answer rather than by how well it grades, and a leaderboard that reported only the median would hide that. We report the second-smallest value alongside the smallest on both tasks, and the abstention rate alongside both, so that a reader who wants the median has it and a reader who wants to see what sets the bottom of the field can see it.

Finally, the aggregation convention of §5.3 leaves the leader-to-tail difference unchanged on all four tasks, to the reported precision, so the margins above do not depend on it.

5.5 The second chain under a null, and what its positive reading is worth

The claim of §4.2 about the human-gold chain — that its positive token names a comparison rather than a verdict — is easy to assert and hard to check against the four margins of §5.4, because those margins have no known truth behind them. We therefore calibrate that chain on a corpus where the truth is supplied outside the pool and the raters can be replaced by ones whose accuracy we set.

The corpus here is our own run — the three judges of §4.2 scoring six systems on 200 TruthfulQA questions — and it is distinct from the released 17-judge panel that §5.7 audits. Its external truth is the answer table released with TruthfulQA (23): each of the 200 questions ships a list of correct and a list of incorrect answers, both written by the dataset's authors. A deterministic lexical rule of ours — content-word overlap with a polarity guard, no model in the loop — maps each of the 1,200 (question, system answer) pairs to positive, negative, or undecidable. It resolves 608 pairs (75.00% positive) and drops 592, or 49.3%, as undecidable. The rule is where the first disclosure belongs: 251 of the 608 resolved pairs (41.3%) match *both* lists, and are labelled positive by a correct-list-first precedence that we chose rather than derived. Spot checks show these are genuine mislabels — for `tqa019` the answer "Jews have won more Nobel Prizes than Christians" is labelled positive because it overlaps the correct list's "Christians have won more Nobel Prizes than Jews", although it is the incorrect claim. We tried to dissolve the ambiguity with an embedding-similarity rule, but its calibration failed: no threshold pair reached 95% agreement with the lexical rule, and the script fell back to it. We therefore carry the ambiguity as two references rather than resolve it: the full 608 pairs, and the 357 pairs (57.42% positive) that match exactly one list, on which the rule's decision is single-valued. The two differ *only* in how the 251 doubly-matched pairs are treated. Among the six judged systems the unambiguous fraction ranges from 23.5% to 35.0% of each system's 200 answers (33.1% to 74.3% of its resolved pairs), so the difference between the references is not a random subsample. No actor in the pool — judge or judged — can affect any label.

The positive reading is reachable, and on the wider reference it is taken by the rater with no ability. Adding three raters that have no judging ability at all — one that answers positive on every item, one that answers negative on every item, and a fixed-seed random rater — to the two judges that cover all 608 pairs, the leader is `baseline_always1` at 75.00%, which is the base rate itself; the median rater is at 64.64%, so the margin is +10.36 points and the chain returns ABOVE_MEDIAN_ACTOR. The two real judges sit at 69.08% and 64.64%, both *below* the rater that answers positive on everything. The verdict is therefore not the negative result one might have expected — it fires — but what it fires on is the reference's base rate.

The identity of the leader is a property of the reference, not of the raters. Keep the same actors and the same votes, and change only the treatment of the 251 doubly-matched pairs: dropping them leaves 357 items, and the leaderboard becomes Qwen3.8-27B at 82.07% and DeepSeek-V4.1-Flash at 81.51%, against the positive-everywhere rater at 57.42%, the random rater at 56.86% and the negative-everywhere rater at 42.58%. The median is 57.42%, the margin +24.65, and the token is again ABOVE_MEDIAN_ACTOR — this time with the Holm correction in the same report rejecting 3 of 4 comparisons, where on the wider reference it rejects 2 of 5. The three-judge version is limited only by coverage: on the 206 items that the third judge has answered, the three judges sit at 84.95%, 83.98% and 83.50% and Holm rejects 3 of 5. One change of labelling convention, and the rater that answers positive on every item goes from first place to third-from-last. This is the paper's own thesis applied to a corpus we did not construct: on this corpus the reference, not the rater, decides the leader. The design cannot separate ambiguity in the gold labels from a matcher that is reliable only on the items it can decide — the unambiguous set is selected by the same rule — so we claim the first as the operative quantity, not as the excluded cause of the second.

Against that, the chain's behaviour when there is nothing to find is the number a reader needs before quoting the token. We built 80 independent pools of six synthetic raters whose true accuracies are identical (85% against the same pair-level gold), so that any difference the chain reports is a difference it invented. On the unambiguous reference it returned ABOVE_MEDIAN_ACTOR in 42 of 80 pools, a null false-positive rate of 52.5% with a 95% interval of [41.7, 63.1]; on the wider reference the same construction gives 35 of 80, or 43.8% [33.4, 54.7]. The mechanism is not a subtle one: the branch is the point comparison margin ≥ 2.0 points with no interval attached (`audit.py`, the verdict branch), so as long as the leader of a finite pool is an order statistic, its excess over the median is positive on average and a threshold of two points is crossed

by sampling noise alone at this rate. The null pools' mean margins, 2.25 and 1.87, sit on that threshold. The chain's own resolution is coarser than its threshold, which we measured by setting the gap deliberately: Table 5 sweeps the true accuracy gap between one rater and the other five, on the unambiguous reference. The 50% point of the chain falls between a 0- and a 1-point true gap, 80% is reached only at 4–5 points, and 100% at 8 points and above. A declared threshold of 2.0 points therefore sits in the rising part of the curve, below the 80% level, which is the quantitative form of the complaint.

Gap between leader and others (pp)	Repetitions	`ABOVE` returned	Mean margin (pp)
0 (true null)	60	28 (47%)	2.25
1	25	17 (68%)	2.82
2	25	17 (68%)	2.59
3	25	18 (72%)	3.06
5	25	22 (88%)	4.16
8	25	25 (100%)	7.23
12	25	25 (100%)	10.97

Table 5: The external-gold chain's operating characteristic, measured on the 357 unambiguous pair-level items with six raters whose true accuracies differ by the stated gap. The 0-point row is the true null. The declared threshold of 2.0 points sits below the chain's resolution: it fires on 47%–72% of pools whose true gap is at most 3 points.

The report already contains the machinery that would withhold the token, and the verdict does not consult it. Each A-chain run prints a Holm–Bonferroni correction over the paired comparisons against the leader. In a 20-pool sub-study of the same null, that correction rejected at least one comparison in 4 runs while the verdict returned ABOVE_MEDIAN_ACTOR in 14; of those 14 verdicts, 10 came from runs in which Holm rejected nothing at all. Verdict and significance agree in 10 of 20 runs, which is what independence looks like. The two are decoupled in the code rather than in principle: the verdict branch reads the point margin only, the Holm block is printed above it in the same report and written to the same JSON record. We report this as a defect of our own second chain, of exactly the kind §4.5 attributes to others, and we do not repair it here — repairing it would change the readings of §5.4, which predate this calibration. The repair we do make is in the direction of the reader: no margin of §5.4 is offered as a verdict, and this section is the number behind that refusal.

One further reading is a warning about quoting any single number from this chain. The same judge scores 77.57% on the 321 pairs that the third judge happened to cover and 69.08% on all 608; on the remaining 287 pairs it is at 59.58%. The gap is not item difficulty, because the excluding half has the *higher* positive rate (77.0% against 73.2%); it is that the covered subset is a partition by judged system rather than a random sample — four of the six systems are inside it and two outside — so a judge's accuracy is stratified by whose answers it is grading. A merged figure is therefore a composition-weighted quantity, and reporting one without the stratum table invites the composition to be read as ability. Every number in this section is in `evidence/positive_control_truthfulqa/`: the two reference files, the two leaderboards, the 60-pool sweeps and 20-pool sub-studies for both references, and the coverage strata, together with the exact label counts of the preceding paragraph. The sweep and pool constructions are `ea_synth_calib.py`, `ea_synth_null_multiplicity.py` and `ea_real_baselines.py`. The scope of all of this is the second chain only: none of it moves a leave-one-out reading of §5.1, §5.2 or §5.6, whose floor is a different quantity computed against a maximum over generators rather than a median over raters.

5.6 Twenty-five generators: the ruler fires, and what it returns

The two judge-gold corpora above have exactly two generators, which is the case Proposition 1 predicts, and the human-gold corpus has no model-generated gold at all. To measure the ruler in the regime it was built for we keep the same annotation set of §5.4 and change only the gold: instead of the human labels we take the majority vote of the 25 model judges themselves, and we rank those same models against that gold. This is the situation the title names, the label producers competing on the board they defined, and three things change at once. The floor becomes a self-reproduction rate rather than an agreement rate (self_reproduction,

25 generators against a minimum of 3). Nothing in the construction forces the two sides onto one item set; it happens on three of the four tasks because each of the 25 generators leaves a reduced table behind, and on those three the floor and the leaderboard are computed over one shared item set (same_domain_as_floor in `evidence/m25_summary.json`), where the two-generator corpora could not be aligned at all. We note what that alignment does not buy: $\delta = 2.0$ never gets to bind on this construction, and Proposition 4 says it cannot bind on any construction shaped like it.

What it returns is a wall, not a tier. On all three decidable tasks the floor now coincides with the leader's own score: factuality 93.83% against 93.83%, coherence 81.12% against 81.12%, relevance 96.86% against 96.86%, each a margin of 0.00 points. Preference shows the same equality (97.2603% against 97.2603%, and the report prints the margin as 0.0 before it refuses), yet it is not readable either, and for a different reason: the domain gate, not the floor, stops it, because two of its systems, `claude-opus-4-7` and `claude-sonnet`, are scored on item sets that differ from the floor's and Algorithm 1 refuses a board whose rows do not share a denominator. The intervals agree with the point estimates rather than rescuing them: the paired case bootstrap puts factuality at $[-3.90,\ 0.00]$, coherence at $[0.00,\ 0.00]$ and relevance at $[-0.71,\ 0.00]$, so on none of the three does the optimistic tail of the resampling lift the leader above the floor it helps define: the leader is not merely within δ of the floor, it is the floor. The interval is not identically zero on two of the three tasks, and the reason is in the per-draw construction rather than in a moving domain. Each round draws one case resample and scores both arms on it; the leader's column, however, is computed against the full gold table while the floor's column is the best leave-one-out score among the generators, the leader included, and that maximum is re-taken inside each round. The interval is therefore a statement about a fixed leader against a re-maximised floor; it is not an interval for the identity, which is exact at the point estimate and needs none.

At the point estimate the imposed-gold construction makes the two equal to the last decimal; in a resample the re-maximised floor can sit above the leader's own column, so the lower endpoint is negative on factuality and relevance while the upper endpoint is zero on both — the asymmetry follows the direction of the max operator, and it does not put the leader above the floor. Coherence is the case where no round separates the two, which is why its interval collapses. Figure 1(b) draws those three rows with their intervals; its other panel is the rank exchange between the two gold definitions of §5.2. An earlier version of this section reported MARGINAL on coherence at +1.6037 points [0.5276, 2.9412] and MARGINAL on relevance at +0.1023; those readings are withdrawn in Appendix A, and the reason is the first of the two disclosures below.

The mechanism is worth stating because it is this paper's claim reduced to one line, and because, we note, the honest form of it is a proof rather than an observation. The leader is one of the 25 models that produced the gold. Removing one generator's vote can never change a plurality label, only vacate it, and we note why: the count of one answer falls by one and no other rises, so the holder of the largest count either keeps it outright or meets a tie and loses its label. The carrier states the same thing as a count over all 25×4 generator-task cells, and the count is 0 (`label_change_pairs`, `evidence/m25_selfbias.json`), which is what Proposition 4 predicts rather than what a sample would estimate. What removal does cost is items: a generator is decisive, in the sense that its own vote is the reason an item carries a label, on between 2 and 25 items — the extremes over those cells, not a spread within one corpus, whose medians are 7, 19, 4 and 5. So each generator's leave-one-out self-reproduction is its own board score on the items that survive, and the maximum over generators — the number a leader must clear — *is* the best generator's score. Scoring on the shared item set of 486, 445, 987 and 511 items per task, the ruler therefore returns no verdict better than UNREADABLE on a judge-generated board once the gold rule is stated honestly; internally the ranking is still resolvable (8, 0, 4 and 2 of the 24 case-clustered comparisons against the leader are statistical ties), which is precisely the reading a citation must not trade on. A four-fold floor, which removes six or seven ballots at once, moves three of these four readings (factuality and relevance into the marginal tier, coherence into the readable one at +3.96 $[+2.19, +5.84]$); we report that as a bound on the construction rather than as a second result, because it estimates a different floor and not a better one (`evidence/strict_review_followup/EC63_kfold_vs_loo.json`).

Both denominators are printed here, because the identity of Proposition 4 is a statement about one of them and a reader has to be able to tell which column a quoted score came from. On the unreduced gold the

same four leading systems score 93.17, 78.44, 96.59 and 96.35 percent, and those are not the numbers the floors above are set against: the reduced convention does not keep a label on 12, 42, 9 and 9 of that gold's items, and every floor and margin reported in this section is computed on the intersection that survives rather than on the unreduced set. The instrument names the column it compared with the registered board, `shipped_on_U`, and its own source records why those two columns are not subtracted from one another: an earlier version of this instrument printed margins built that way, and those margins were readings of nothing on disk. Drop the shared-domain hypothesis of Proposition 4 and the equality stops holding, by the distance between the two columns just named; that distance is the price of the hypothesis, and it is the reason the rule refuses a corpus whose two sides do not share an item set instead of reporting a margin across them.

The second disclosure is the audit of the rule itself. The tie test was originally $\max(c) \cdot 2 = \sum(c)$, which recognises only an exact two-way split: on 2, 13 and 4 items per task (factuality, coherence, relevance) the largest vote count was shared while a third answer held the remainder, and the gold for those items fell to the lexicographic order of the answer strings — the rule this paper refuses to adjudicate with — while the same test voided 5 items of the form 4/2/2 that do have a unique plurality winner. The criterion is now stated as it reads: a gold label exists on an item iff the largest vote count has a unique holder, and 0 items anywhere in this corpus have their gold decided by string order. Separately, the floor entered the margin after rounding to two decimals while the leader did not, so coherence's +1.6037 contained a component that its own interval contradicted; ceilings and margins are now compared at full precision, and the equality case that this exposed is judged as the declared rule says — $\max_s \mathrm{acc}_c(s) > \mathrm{floor}_c + \delta$ is strict, so a margin of zero has not cleared anything and reads UNREADABLE.

That leaves the question this construction was built to answer, which is whether being a label producer flatters a model. It is also the quantity Proposition 4 leaves uncollapsed: the margin is pinned at zero, but the two sides of that identity can both move. For each of the 25 systems we recompute its board score with the items that only its own vote decided removed from numerator and denominator together. The sign of that shift is not a finding, and we say so before giving any number: an item is decisive for a generator only where that generator's own answer is the unique plurality winner, so the generator is correct on every decisive item by the definition of the rule it is scored under, and removing rows that are all correct cannot, under the rule in this paper, raise the score they were taken from. The carrier states the same fact as a field rather than as an argument: its per-generator hit rate on decisive items is 100.0% on all 100 generator-by-task rows, and the instrument's own self-test asserts both that boundary and that a generator with no decisive item has a shift of exactly zero. What the shift can carry is therefore its magnitude, and the magnitude is what the overlap is worth: medians of +0.18, +1.60, +0.03 and +0.10 points and leader-specific shifts of +0.11, +0.97, +0.02 and +0.03.

So the zero margins above are not a coincidence at the top of the distribution: each is a leader sitting at a floor that its own decisive votes helped raise, and removing them puts every arm on the negative side. Both readings are in `evidence/m25_selfbias.json`, whose own cross-check requires its recomputed board scores to match the registered leaderboard within the tolerance that file declares (0.01 points) before it will write anything. A quantity whose sign the rule fixes cannot, in this paper, separate flattery from reproduction. What the shift sizes is the number of correct items each generator handed itself, and the mechanism a reader will supply first — judges from the same family agreeing because they were trained on overlapping data rather than because either reproduced the labelling convention — would show up in that count and in the permutation screens, not as an opposite sign. Testing it needs a contrast the identity does not force, such as scoring each system against gold built with its family-matched peers held out, and that measurement is not in this paper (§8).

The tie rule is load-bearing from the other side, and we measured by how much. The rule this paper reports leaves tied items without a gold label, and a rule that adjudicates ties instead puts them back in, which breaks the identity that pins the margin at zero: re-running the three decidable arms with the tied items admitted — uniformly at random over the tied answers under ten seeds, which agree on every arm, and with each vote weighted by its judge's own human-gold macro on the same task — returns a marginal verdict on factuality and relevance under both rules and a readable one on coherence, whose margin stays above $\delta = 2.0$ in all ten seeds (between 2.2 and 4.0 points) and reaches 3.2 under the weighted rule, from 2, 13 and 4 tie items per task. The flip is not evidence of readability; it is Proposition 4's identity losing its hold one

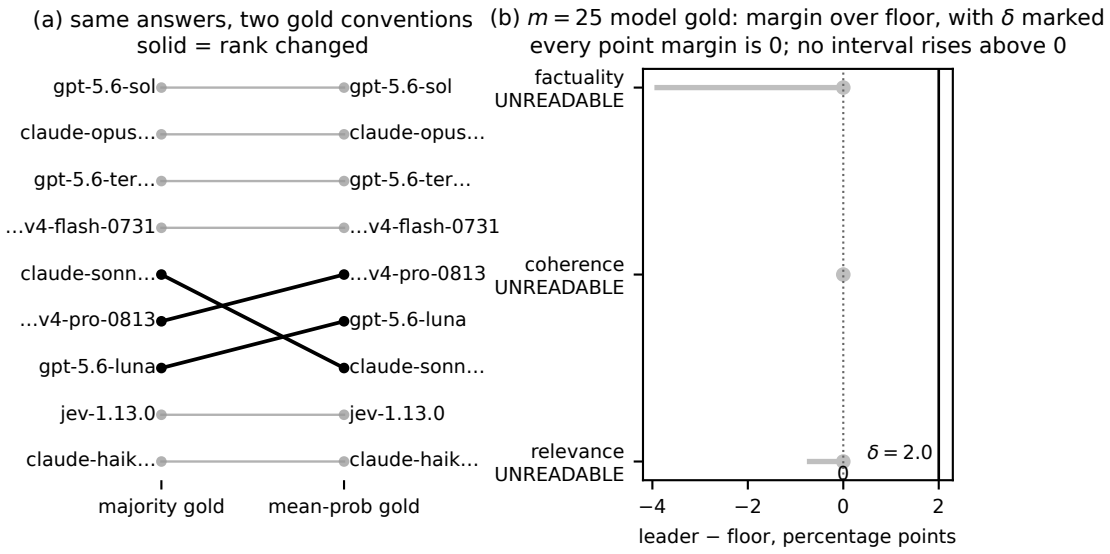


Figure 1: Two shapes the tables leave out. Panel (a): the rank of each of the nine contestant systems on the 14,292-question corpus under majority gold and under mean-probability gold, both read under the tie rule that resolves disagreements by a fixed order; solid segments mark the three systems whose rank moves, and the leader holds first place under both gold rules. Panel (b): the leader's margin over its own leave-one-out floor on the 25-generator construction of §5.6, with the 95% paired-bootstrap interval and the declared $\delta = 2.0$. Every row is instructive in the same way once the gold rule is stated honestly: each point margin is exactly zero and each interval has zero as its *upper* endpoint, so not even the optimistic tail of the bootstrap puts the leader above the floor it helps generate — the ruler does not merely fail to clear δ, it fails to clear 0. Preference is absent because that task is stopped by the domain gate rather than by its floor: its margin is also 0.0, and the report prints that number before refusing, so drawing it would add a fourth instance of the identity of Proposition 4 rather than a fourth reading. System names on the vertical axis are shortened to the panel's width; the full identifiers are the keys of the released reports. Both panels are generated from the registered carriers by `make_figures.py`; every plotted value is read from a registered carrier.

convention at a time, and a reader citing this construction is citing a margin that three choices hold at zero, of which the tie rule is one. The preference arm has no tie items and does not move; the question corpus's verdict is unchanged under random tie-breaking; the decision corpus's refusal changes class, from domain mismatch to floor degeneracy, without becoming readable; the human-gold chain has no vote gold, so this lever does not exist there (`evidence/strict_review_followup/EC32_tie_break_sensitivity.json`).

5.7 The same reading on a panel we did not build

The construction of §5.6 is ours, and a reviewer may reasonably read its result as an artefact of how we imposed the gold. This section puts that reading to a test on bytes we did not assemble, pointing the same ruler, without changing it, at a panel whose votes somebody else published: the per-judge release described in §3, 17 judge models scoring 5,918 TruthfulQA items, one prediction file per judge, each row carrying that judge's verdict. Gold is the panel's own majority under the plurality rule of §4.1; the ranked field is the same 17 models against that gold. The input is assembled by `make_panel_votes.py` and audited by the same `audit.py` that produced every other reading here, at 2,000 paired bootstrap rounds and the seed used throughout.

What comes back is `UNREADABLE` at a margin of exactly 0.00 points. The floor is 91.4993%; the leader, `OpenHermes-2.5-Mistral-7B`, scores 91.4993%; the paired interval of their difference is $[-0.744,\ 0.00]$, whose upper end is zero, so no resampling of this panel puts the leader above the level it must clear. Sixteen of the

17 systems score strictly *below* the floor, the next of them, `Starling-LM-7B-alpha`, at 91.369%, and the panel's own spread across the field, measured on the full gold rather than on the shared leave-one-out domain, is 44.37 points, so this is not a board collapsed by near-ties: it is a board whose top sits exactly at the level its own top generator reproduces. The item sets agree as required — the floor and the leaderboard are both computed on the same 5,376 items, out of 5,918, the 542 excluded ones being those where removing a judge leaves the gold without a unique plurality holder — and we report the gold's own coverage as complete: every item carries a label, so the items excluded from the shared domain are excluded by the leave-one-out rule rather than by a missing label, which is the difference from the refusal of §5.1.

Two disclosures travel with the number, and neither is a hedge. First, the ranked field *is* the generating pool: all 17 actors appear as judges and as contestants, which is the configuration §5.4 calls self-grading, so the reading bounds alignment with the panel and says nothing about favouritism. What this section establishes is narrow and worth stating exactly: Proposition 4's hypotheses hold on bytes we did not assemble, so the zero margin of §5.6 is not an artefact of our table-building step — the identity is instantiated, at the same margin of zero, on votes we did not produce. It is not a second measurement of a collapse, and it says nothing about *this* panel in particular: the deduction needs only that the field be the pool, and this release ranks the same 17 models whose predictions make its gold. What a reading here *could* establish, and what remains unestablished, is the disjoint case the ruler was built for; §5.9 measures that configuration on a separate release and reports the refusal it returns, and §8 keeps the scope limit that a disjoint board returning a positive verdict is still unmeasured. Second, this is the one reading in this paper that we report as not reproducible offline: the release declares no licence, so we cite its votes instead of redistributing them, and the shipped artifacts are the derived report `evidence/panel_truthfulqa.json`, the per-file digests `evidence/panel_votes_digests.json`, and the folded summary `evidence/panel_summary.json`, from which `rerun_panel.sh --verify` answers whether a given log was produced by the code now on disk.

The floor's definition is itself one of those conventions, and the sensitivity question it raises was put to us directly: does any verdict here depend on taking the pool's hardest generator as the floor rather than a median one? We recomputed every arm of §5.4, §5.6 and this section against a median floor, the median re-taken inside every bootstrap draw and both arms of each contrast sharing the same draws (`evidence/strict_review_followup/EB_leader_vs_median.json`). On the m25 construction the substitution un-pins the identity: the leader-minus-median margin is +5.56 points (interval $[+2.45, +8.49]$) on factuality, +19.33 ($[+14.61, +23.31]$) on coherence and +6.69 ($[+3.66, +7.89]$) on relevance, every interval excluding zero, so a median floor would return a readable verdict on all three; preference keeps its refusal under either floor, its domain gate firing before any floor comparison (the margin it would print is +6.46, $[+4.30, +8.76]$). The direction is the finding: the leader sits inside the pool that median is taken over, so a positive leader-minus-median is the self-grading signature Proposition 4 predicts, not evidence that the labels reproduce — the flip changes which verdict a convention returns, not what the construction supports, and these counterfactual readings are computed under a different rule than (3), whose floor is the maximum over generators. On the human-gold chain the substitution is vacuous rather than reassuring, because that chain's floor basis already is the median actor (§5.4); the non-trivial contrast there is the leader against the hardest annotator, and that difference is 0.00 points on all four tasks — the leader is the maximum of the field, and the interval's upper end is zero on each. On the two-generator corpora the pool's median equals its maximum, so the lever does not exist and the refusals stand under either reading. On this panel the contrast is +6.36 points at the point estimate, and we print no interval for it: the votes are cited rather than redistributed, so the paired draws cannot be recomputed, and the per-generator values the median is taken over are rounded in the carrier that reports them.

The pool's composition is a choice of the same kind, and it gets the same treatment: six further judges — free-API models served by Token Factory, run under the original JudgeSense prompts on a 100-item sample drawn per task — merged into the human-gold chain grow it from 25 judges to 31, and on the three tasks whose shared domain is the new judges' full sample every verdict of §5.4 stands with margins moving by at most one point; none of the 24 single-judge ablations, one per new judge per task, flips a verdict; on preference the merge decides nothing, because the 31-judge common domain collapses to 12 items, at which granularity a single item moves the task macro by more than four times the declared threshold (`evidence/strict_review_followup/EC57j_pool_composition.json`).

5.8 Calibrating the floor against a ruler outside the ranked field

The panel reading of §5.7 measures the floor against the panel's own gold, and on that reading certification is blocked because the ranked field is the gold pool — a property of who generated the gold, not of the measurement. One released panel comes with a ruler that stands outside its actors: the same release that publishes the votes publishes, alongside each prediction, the item's own benchmark label (§5.7 describes the release; its license situation is recorded there and applies here unchanged, so the ruler travels as digests and derived carriers, and the per-item labels are not redistributed). That makes a preregistered variant of the measurement possible, registered before it was run (`evidence/calibrated/PREREG.json`): split the items into a calibration subset C of declared size and an evaluation remainder E; score every actor a against the panel's own gold exactly as in §5.7, giving s_a; measure on C the per-actor bias $\hat{b}_a$ between that score and the actor's agreement with the ruler; and on E replace s_a by the debiased score $\hat{t}_a = \text{mean}_E s_a - \hat{b}_a$, with standard error $\sqrt{\text{var}_E(s_a)/|E| + \text{var}_C(s_a - t_a)/|C|}$. The floor and the leader are then compared on $\hat{t}$ in the form of (3), the bounds are simultaneous across the m judges and k contestants (normal quantiles at $1 - \alpha/2m$ and $1 - \alpha/2k$), and certification again requires the lower bound of the leader to clear the upper bound of the floor by more than δ. What the ruler buys is a floor estimate whose bias against an external standard has been measured and removed, not a relabeling of the gold: the ranked field is still self-graded, and the ruler enters only through the calibration split and through the scoring of the certifications.

On 9,600 resampled boards of the same panel configuration as §5.7 ($m \in \{3, 5, 7, 10\}$ judges, $k \in \{3, 5, 7\}$ contestants, $|C| \in \{50, 200, 400, 800\}$, 200 draws per cell), 32 fail the estimator's minimum-domain guard; on the 9,568 measurable boards the corrected rule certifies 1,242 (13.0%) and the uncorrected rule certifies none. The correction does what it is supposed to do at the level of the estimate: the median per-actor bias reduction against the ruler is 0.93 across all boards, and the estimated margin tracks the measured external margin (means 5.8 and 5.9 points). The preregistered primary cell ($m = 7$, $k = 5$, $|C| = 400$, $\delta = 0.05$) certifies 6 of 200 boards (3.0%); the certified boards all name the external top (false-certification rate 0.000, Wilson [0.000, 0.390]); the debiased rule's top equals the external top on 0.900 of the scored boards against 0.025 uncorrected; the board-level separation of correct top-naming by the estimated margin is AUC 0.686 [0.530, 0.831]; the median bias reduction is 0.918. The cell's true external margin is -2.2 points — its best debiased judge is at least as accurate as its best contestant — so the correct reading is silence, and 194 of the 200 boards return it. The preregistered verdict for this cell is the registered weak zone: the bias-reduction criterion is met, the false-certification point estimate is met while its interval cannot tighten on six certifications, and the AUC criterion (≥ 0.70, lower bound above 0.60) is not met.

Certification is real where the configuration leaves a margin to certify. Table 6 lists the cells whose true external margin is positive: on those eight cells certification rates run 22.5–65.0%, the false-certification rate runs 0.000–0.069 with Wilson upper bounds at or below 0.151 (one cell rounds to the registered 0.15 threshold and is reported as measured), and AUC runs 0.710–0.846 with lower bounds 0.592–0.730, one of which sits below the registered 0.60 floor. The $m = 7$ and $m = 10$ cells certify in at most 8.0% of boards: with seven or ten judges in the floor's pool, the panel's strongest judge matches or beats its strongest contestant against the ruler, and the corrected rule declines to certify what is not there. At $\delta = 0.02$ the same stored bounds re-certify four cells at 4.5%, 57.5%, 69.0% and 25.5% with false-certification rates 0.000, 0.043, 0.036 and 0.000.

Two limits are disclosed with the arm rather than absorbed. First, the calibration bias is itself an estimate: a split-half transfer check, which estimates the per-actor bias on one half of C and applies it to the other half, runs a median absolute residual against the ruler of 5.6 points at $|C| = 400$ and 4.0–4.4 points at $|C| = 800$, shrinking with $|C|$ as a finite-sample estimate should; the registered 4-point failure threshold is crossed at face value on most cells, and we report the arm as the registered weak zone rather than as a clean pass. Second, the quantity certified is not the quantity §5.7 could not certify: it is the debiased alignment of a self-graded field, with the ruler entering through calibration and scoring only. On semi-synthetic boards with a planted truth, the same estimator recovers the floor with median bias reduction 0.84–0.92 when the judges' errors are correlated ($\rho = 0.5, 0.9$) and 0.29 when they are independent ($\rho = 0$) — the case where the uncorrected panel gold is already aligned with the truth (its uncorrected top-naming is perfect in that cell) and there is little bias left to remove; the estimator's self-test recovers a planted per-actor bias of ± 0.15, induced by a planted 15% gold–truth disagreement rate. §5.9 measures the remaining case — a leaderboard

whose contestants are disjoint from its gold pool — on a separate release, and finds it returns a refusal rather than a certification.

m	k	$\|C\|$	true marg. (pts)	$\hat{\Delta}$ (pts)	certified	FCR [95% W]	AUC [95% boot]	uncorr. top
7	5	400	-2.2	-2.5	3.0%	0.000 [0.000, 0.390]	0.686 [0.530, 0.831]	2.5%
3	3	200	21.8	22.3	26.6%	0.038 [0.010, 0.128]	0.723 [0.633, 0.810]	14.7%
3	3	400	21.7	21.9	40.5%	0.000 [0.000, 0.045]	0.784 [0.717, 0.845]	14.6%
3	3	800	21.7	22.0	50.5%	0.020 [0.005, 0.069]	0.710 [0.592, 0.812]	14.6%
3	5	200	29.4	30.2	36.9%	0.069 [0.030, 0.150]	0.739 [0.631, 0.836]	6.1%
3	5	400	29.0	29.3	51.5%	0.049 [0.021, 0.109]	0.766 [0.651, 0.869]	7.0%
3	5	800	29.0	29.6	65.0%	0.038 [0.017, 0.087]	0.781 [0.655, 0.897]	6.6%
3	7	400	30.9	31.3	55.0%	0.054 [0.025, 0.114]	0.719 [0.606, 0.818]	3.0%
5	5	800	6.1	6.0	22.5%	0.000 [0.000, 0.079]	0.846 [0.730, 0.950]	6.1%

Table 6: The calibrated-floor arm on the released panel. Each row is a cell of the preregistered grid (m judges, k contestants, calibration size $|C|$; 200 resampled boards; $\delta = 0.05$); the first row is the preregistered primary cell. True marg. is the measured external-ruler margin between the best contestant and the best judge, in points; $\hat{\Delta}$ its debiased estimate. Certified is the share of boards the corrected rule certifies; FCR the share of certified boards naming the wrong external top (Wilson 95%); AUC the board-level separation of correct top-naming by estimated margin (bootstrap 95%); uncorr. top the uncorrected rule's share of boards whose raw top equals the external top (its certification share is zero in every cell). Regenerated from `evidence/calibrated/summary_ci.json`, recomputed from the per-board carrier by `make_calibrated_summary.py`; the raw boards are `evidence/calibrated/e2_truthfulqa_boards.csv`.

5.9 A disjoint-field deployment on AgentJudgeBench

The panel of §5.7 has its contests among its judges; the calibrated arm of §5.8 corrects the estimate but leaves that identity in place. A preregistered deployment on a panel whose contestants are disjoint from its judges tests whether the same instrument certifies where the configuration leaves a true positive margin to certify. The corpus is (36) (AgentJudgeBench), a public release of 48,522 (task, generator-output) items scored by seven judge models and five generator models, with the item's own programmatic score embedded per record. The primary configuration uses six judges and five contestants sharing no members (`gpt_5_4` dropped from the judge pool so that judges $\cap$ contestants $= \varnothing$); secondary arms widen the judge pool to seven, add `gpt_5_4` back as a judge, or drop prometheus2. Votes are binarized strictly: a judge votes 1 iff all four metric scores equal 1.0, the truth is 1 iff the embedded programmatic score equals 1.0, and the paired bias $\hat{b}_a$ is defined on C against the external ruler as in §5.8. The grid is $|C| \in \{200, 400, 800, 1600\}$, 200 resampled boards per cell at $\delta = 0.05$, preregistered at `evidence/calibrated/PREREG_P4.json`.

On 3,200 boards across the four arms the calibrated rule returns `UNREADABLE` or `MARGINAL` and never `READABLE`: 0 certifications (3,020 `UNREADABLE`, 180 `MARGINAL`). On the primary arm the `UNREADABLE` share is 83.5% ($|C| = 200$), 86.5% (400), 95.0% (800) and 99.5% (1600), the remainder all `MARGINAL` (729 of 800 overall). The uncorrected rule's point margin is positive on 100% of primary boards (+1.46 pts mean), and the split-half transfer check runs 2.6–7.3 pts at $|C| = 200$–400 shrinking to 2.6 pts at 1600. At $|C| = 1600$ the primary cell has 199 of 200 boards `UNREADABLE`. The estimator recovers each generator's true rate within a median of 1.6 points at $|C| = 1600$ (per-generator 1.1–2.2 pts), and names the true best contestant on 94.4% of all 800 scored boards (V1 84.0% at $|C| = 200$, 95.0% at 400, 98.5% at 800, 100.0% at 1600, Table 7). The raw rule also names the true argmax but with a positive margin (+1.46 pts) where the true gap is −7.1 pts, so certification rather than ranking is what fails in this setting.

On the cells with a positive true external margin (the $m = 3$, $|C| \geq 400$ cells of Table 6), the certified arm remains the self-graded panel of §5.8. The preregistered criteria (H1: silence on $\geq 95\%$ of boards at $|C| \geq 800$ in every arm and 0 `READABLE` throughout; H2: recovery ≤ 0.05 at $|C| = 1600$; H3: V1 ≥ 0.95 at $|C| \geq 800$) all hold; the strict H1 clause is met at $|C| \geq 800$ and as `MARGINAL` rather than `UNREADABLE` at $|C| \leq 400$, which does not trigger the registered failure mode because `READABLE` never fires. We report the criteria as met in this setting.

$\|C\|$	true	$\hat{\Delta}$	READABLE	UNREADABLE	MARGINAL	V_1 (%)	AUC	uncorr.	placebo
1600	-7.1	-6.4	0	199	1	100.0%	—	+1.46	2.6
800	-7.1	-6.4	0	190	10	98.5%	0.973 [0.939, 1.000]	+1.46	3.7
400	-7.1	-6.1	0	173	27	95.0%	0.911 [0.860, 0.951]	+1.46	5.2
200	-7.1	-6.5	0	167	33	84.0%	0.725 [0.616, 0.820]	+1.46	7.3

Table 7: The disjoint-field deployment on AgentJudgeBench (§5.9). Each row is a preregistered cell (m judges, k contestants, $|C|$ items; 200 resampled boards at $\delta = 0.05$, primary pools). The judge–contestant pools share no members; the true margin is the best contestant minus the best judge against the embedded programmatic truth. *uncorr.* is the same point margin computed without the calibration; *placebo* is the transfer check, the median absolute residual left when the correction is fitted on one half of the item pool and applied to the other. All margins are in points; AUC is printed with its 95% bootstrap interval and is undefined where a cell has no "wrong top" to separate. Regenerated from `evidence/calibrated/p4_ajb.json` and `p4_ajb_withgt.json`, computed from the per-board carrier by `calibrated_floor.py`.

5.10 The latent-variable floor is not measurable on this corpus

A third ruler is available in principle: fit a latent-variable model to the annotators themselves rather than to the raw majority, and ask what floor the fitted truths imply. We fitted it, and on this corpus it does not yield a floor we are willing to report. What follows is therefore a negative result about the estimator and about these annotations, not a measurement of the floor, and we say so before giving any number so that no value below is cited as one.

The model is the usual one (10). With $z_i \in \{1, \dots, K\}$ the unobserved truth of item i, a prior $\Pr(z_i = k)$ over truths, and a per-annotator confusion matrix $\pi_j[k, v]$ giving the rate at which annotator j reports v on an item of truth k, the complete-data log-likelihood is $\sum_i \log \sum_k \Pr(z_i = k) \prod_j \pi_j[k, v_{ij}]$, which the code evaluates by summing within an item before taking the logarithm, because taking a logarithm per observation would assume the truth is known. The fit alternates an M step that forms $\Pr(z_i = k \mid \text{obs}) \propto \Pr(z_i = k) \prod_j \pi_j[k, v_{ij}]$ and an E step that re-estimates $\pi_j[k, v] = \big(\sum_i \Pr(z_i = k \mid \text{obs}) \mathbb{1}[v_{ij} = v] + \alpha\big) / \big(\sum_{v'} \sum_i \Pr(z_i = k \mid \text{obs}) \mathbb{1}[v_{ij} = v'] + \alpha K\big)$ with $\alpha = 1.0$, and stops on the largest change in any entry of π falling below 10^{-6} after at most 200 iterations, with 8 initialisations drawn from seed 20261005 and the screens that Table 8 reports as *rej* discarding n_{init} of them before the fit is accepted; the run converges at iteration 20, 32, 16 and 10 on the four tasks, so the tolerance reported here is 10^{-6} for the arm that produces §5.10's table and 10^{-10} with a cap of 500 iterations for the sweep of §5.12, which is a different tolerance because a synthetic corpus with a known truth is the one place it is checkable. Two properties of that estimator decide what follows. The first is the independence it assumes: the product over annotators treats their errors as conditionally independent given the truth. The second is the smoothing that this dependence makes non-optional: with $\alpha = 0$, an annotator that omits a label entirely sets that cell to zero and the M step then zeroes the posterior of every item carrying it, so the fitted truth becomes a function of the initial values rather than of the annotations.

We report the estimator in this form so that the numbers below can be read as what the estimator, not the data, selected. Fitting the latent-variable model to the same 25-annotator annotations and comparing the weakest annotator's agreement with the fitted latent truth against its agreement with the pool's majority gold gives a signed quantity, and the sign is not constant. The four pairs, as (majority, latent) in percent, are $(72.40, 71.60)$ on factuality, $(39.00, 31.40)$ on coherence, $(88.41, 88.41)$ on relevance and $(55.56, 58.33)$ on preference, so the signed difference $\Delta = \text{latent} - \text{majority}$ is -0.80 and -7.60 points on two tasks and $+0.00$ and $+2.77$ points on the other two. The positive case is the informative one: it means the model has recoded which annotators agree with the majority, so an annotator that looked unreliable under majority vote is not unreliable under a model of annotator behaviour. These four numbers are not read off the estimator alone, and the table reports what had to be true for each of them to exist. The estimator is started from eight initialisations, one deterministic and seven driven by a seed, and each is screened before any of them is allowed to define the reported agreement. A screen rejects a solution that promotes a label which fails to win a majority on any item, because such a label is unsupported as a latent truth: on the factuality task

one label is the median answer on no item of the 500, and on the items where it appears at all it falls short of half the votes. A second screen rejects solutions the likelihood does not distinguish from their own label permutation; permuting the latent labels leaves the observed likelihood unchanged to machine precision, and on the preference task the eight initialisations agree on the likelihood to within 10^{-5} while their latent truths are permuted copies of one another.

That second screen is the one that makes the result what it is, and it is worth being precise about why it exists: the log-likelihood of a latent-class model is invariant under a relabelling of the latent classes, so a likelihood is not a selection criterion among permuted solutions. Concretely, in a two-label fit on one task two starting points agree on the log-likelihood to within 3×10^{-6} yet one agrees with the majority label on 0.994 of items and the permutation-twin on 0.006, so the two are the same likelihood read two ways, and the choice between them is worth points rather than a footnote: the reported value is the pool's agreement with whichever of the two is picked, and the *anchor* column of Table 8 reports that agreement. A rule that selected the solution by likelihood alone would therefore be selecting uniformly among solutions that disagree about the answer, and the choice among them would be the initialisation's, not the data's. On two tasks the screens reject every random initialisation, leaving only the deterministic one; the table records how many of the eight survived. Two tasks go further. On the relevance and preference tasks the minority label is removed before the fit, because no starting point yields a usable solution with it present—five different seeds converge to the same basin—so the reported agreement is a property of the two-label space, and the table names the label that was removed. On the coherence task the surviving solution agrees with majority vote on only 0.836 of items, against 0.988 to 0.994 on the other three; that is the anchoring strength that resolves the permutation ambiguity, and at 0.836 it is the weakest of the four. Taken together these conditions say that the four values in Table 8 are not four estimates of one quantity, and that none of them is an estimate of a floor in the sense (1) defines.

What the exercise establishes is negative and worth stating plainly, and it is a narrower statement than an identifiability claim: on 500 to 1,000 items from 25 annotators, the fit is permutation-ambiguous on exactly the two tasks whose Δ is not negative — the screens return `permutation_ambiguous` true for those two and false for the two whose Δ is negative — and the case-level resampling of the next paragraph reproduces the sign on the negative pair and fails to reproduce it on the other two. Permutation invariance of the likelihood is a property of every latent-class fit and the *anchor* column exists to resolve it; what these four values do not survive is a weak or absent anchor combined with a screen that had to choose between twins the likelihood does not separate. A paper that reported the sign of Δ as a finding would be reporting the outcome of its own screening choices. A signed difference whose sign a resampling does not reproduce is not a measurement of the annotations, so we treat it the way §5.3 treats the aggregators: by resampling the unit in which the annotations actually arrived. Each corpus is organised into cases—two items per case on factuality and coherence, four on relevance, and four on preference, where the four are the original ordering and its swap—so the 500 to 1,000 items contain 130 to 250 cases. We drew 40 cluster bootstrap replicates over `case_id` with replacement, seed 20261011, refitted both golds inside each replicate by the procedure that produced Table 8, and recomputed that agreement on the annotator item-set intersection; the record is at `evidence/ds_case_bootstrap.json`.

On the two tasks with a negative Δ the sign holds on every replicate: the median replicate is -0.935 against a point estimate of -0.80 on factuality and -7.47 against -7.60 on coherence, with 0 of 40 replicates of the opposite sign, and the ambiguity screen flags 3 of 40 and 7 of 40 replicates respectively. On the other two tasks—the same two on which 40 of 40 replicates reported the fit as permutation-ambiguous—the sign does not hold. On relevance the point difference is exactly 0.00, so the reversal is not defined, and we report the 40 replicates as 15 negative, 16 zero and 9 positive across a range of -0.37 to $+0.69$. On preference the median replicate is -3.785, opposite in sign to the point estimate of $+2.77$, and 31 of the 40 replicates are negative or zero. What this adds to the previous paragraph is a measurement of how much of the sign of Δ is a property of the screening and of the resampling: on two tasks it is stable under case-level resampling, and on the two where it is not negative it is not, which are the same two on which the identifiability screens reported the fit as permutation-ambiguous.

An earlier version of this section reported +1.40 and −6.25 points for two of these tasks. Those figures came from a single deterministic initialisation of an estimator whose seed argument did nothing, and they are withdrawn; the detail is in Appendix A.

Task	K_{used}	dropped	rej/n_{init}	minority	anchor	agree(maj)	agree(latent)
factuality	3	—	7/8	UNCLEAR	0.990	72.40	71.60 (-0.80)
coherence	6	—	7/8	1,UNCLEAR	0.836	39.00	31.40 (-7.60)
relevance	2	UNCLEAR	4/8	—	0.994	88.41	88.41 (+0.00)
preference	2	UNCLEAR	4/8	—	0.988	55.56	58.33 (+2.77)

Table 8: Latent-variable agreement against majority-vote agreement, as percentages of the reproduction rate of the weakest annotator in the floor pool, the first of the two columns read against the pool's own majority gold and the second against the fitted latent truth, with the conditions under which each value was produced; the bracketed figure is their difference in points. Neither column is a floor in the sense of (1): that quantity is a maximum over gold generators, while these two are minima over annotators (§4.4), and §5.10 accepts none of the four values as an estimate of a floor. The pool and its common item set are not the reference set of §5.4, so these minima are not comparable with the distributions reported there. K_{used} is the label space actually fitted; *dropped* names a label removed before the fit because no starting point admitted it; *rej*/n_{init} counts starting points rejected by the identifiability screens; *minority* lists labels that win no item majority; *anchor* is the agreement between the fitted latent truth and majority vote, which is what resolves the label-permutation ambiguity. The sign of the difference is not constant across tasks.

5.11 The denominator a convention leaves is not the corpus size

The item set a published score is computed on is not the corpus. Of 14,292 question items, 13,104 are annotated by both generators and 1,188 are not annotated by any generator at all, so item coverage is 91.7%, where a row is one answer by one generator or system on one item, and row coverage is 97.2% (113,837 of 117,069 rows, fewer than $14,292$ items times nine systems because not every system answers every item). Once the convention is applied, the picture tightens further: on 799 of the 13,104 two-annotation items the two generators disagree and the majority rule therefore returns no label, which is 5.59% of all 14,292 items, so 1,987 items carry no gold label and row coverage falls to 91.2% (106,788 of 117,069); after the leave-one-out reduction the common set is 12,406 items. The floor and the two margins of §5.2 are computed on that common set, while each leaderboard there is scored on its own labelled set (11,586 under the default rule and 12,330 under the forced one), and that mismatch of denominators is the reason §5.2 refuses rather than decides; we report the excluded fraction so that the denominator behind each figure can be reconstructed. The decision corpus of §5.1 is a different corpus and is described on its own terms there, including the item counts that no definition reconciles.

5.12 What the estimator can and cannot pin down

One property of the estimator must be stated as a limit on use. We swept it over five random seeds on a synthetic corpus of 150 items with six annotators whose true accuracies span 55% to 95%. The top rank is recovered in all five, and the full ordering in four; in the remaining seed three positions differ and the Spearman correlation is 0.829, which is the adjacent-pair regime rather than a failure to recover the ranking. The absolute values carry sampling error: the estimated span runs from 34.4 to 45.6 points against a true span of 40, two seeds wider than the truth and three narrower, and the top estimate from 89.7 to 96.9 against a true 95. The bottom of the board carries a caveat that the printed aggregate does not: `bottom_estimate_pct` is a minimum *over annotators*, and in the one seed whose ordering is not recovered that minimum belongs to the annotator whose true rate is 71, not to the weakest one — its value there is 57.74 against that annotator's 71, the largest single per-annotator error anywhere in the sweep. Tracked by annotator rather than by rank, the weakest annotator's own estimates run from 49.15 to 59.32 against a true 55; the aggregate's own bottom range, 49.15 to 59.26, agrees with it everywhere except that one seed. Absolute values are therefore recoverable jointly with the ordering, not independently of it, and the

low end of the board is where the two failure modes coincide. The estimator therefore supports statements about the *ordering* of annotators, up to adjacent pairs that this sweep leaves unresolved at 150 items, and its absolute accuracies are recoverable with an error that shrinks as the item count grows. We say so on the strength of a power probe that re-runs the same construction at $n = 150$, 600, and 2400 items: the largest absolute deviation from a known truth falls from 4.84 to 1.25 to 0.70 points and the spread across seeds from 3.75 to 1.45 to 1.06 points, against a binomial standard error that falls from 3.26 to 1.63 over the same range. Our test suite asserts the ordering and the top rank, and stops short of asserting absolute values at a tolerance that a single sweep does not establish. We report the sweep in this form because the previous two versions of this paragraph claimed more than the probe supports; each is listed, with the defect that produced it, in Appendix A. Fixing the seed to drive initialisation, adding the screens described above, and adding the randomness to the sweep itself moved the numbers to the values now given, which are weaker than the earlier claim in the ordering and stronger in the absolute values. What the sweep supports is that the estimated span brackets the truth at this sample size, rather than a claim that the estimate is biased low or that absolute levels are out of reach.

6 The Estimator Is Not the Evidence

The measurements above are only as good as the code that produced them, and an audit tool whose tests stay green under every injected defect would be decoration. We therefore treat our own test suite as an instrument subject to error and measure how much of it can fail. The suite is also the positive control that keeps this section's refusals from being the instrument's only possible output: a construction with four gold generators, where the leader clears the leave-one-out floor by a wide margin, asserts that the ruler returns READABLE (case 8 of `selftest.py`, captured in `evidence/log_selftests.txt`). Without that case the same suite would stay green on a ruler that refuses everything it is given, and every UNDECIDABLE in this paper would be unfalsifiable by construction. What the case does not show is that any real leaderboard is readable: it fixes the generator pool and the margin, which is exactly what the two corpora of §5 do not permit. The unit is a mutation: a single, deliberate defect injected into the implementation, followed by a run of the test suite. A mutation that turns at least one test red is *caught*; one that leaves the suite green is *uncaught*, and we treat uncaught mutations as defects in the suite rather than as acceptable defects in the code. We inject 27 defects into the main auditor, of which the run reports 27 caught, and 15 into the latent-variable arm, of which 11 are caught. Of the auditor's mutations, the caught ones turn between one and forty-five of the forty-six cases red, and two independent guards are shown to be the only thing standing between a defect and a silently wrong interval; the four that survive in the latent arm are initialisations and a redundant guard whose companion mutation does fail, which we report rather than absorb by adding assertions. Both batteries include the measured forms of failure described below, and every count in this section is read from `evidence/log_mutation_audit.txt` and `evidence/log_mutation_ds.txt` rather than asserted; both logs are now written by the run itself, which records its arm's digest before and after and reports a mismatch instead of a result, and a separate read-only check answers the same question in milliseconds without re-running the battery. That check first compared digests taken through the file's *path*, and where a checkout path contains a backslash the hashing tool escapes the printed line: the same clean log then read as stale or fresh depending on which directory the check was invoked from, while the bytes of the code had not moved. The comparison now runs over file contents, and a pinned digest that still carries the escape prefix is refused under its own name instead of being folded into the mismatch verdict, because that verdict would send the reader to restore code that was never touched. Four classes of surviving mutation recur, and each has a lesson attached to it. *Equivalent mutants*: two defensive lines are redundant, so removing either one changes nothing, and only removing both exposes the gap; we measure these with compound mutations rather than declare them acceptable, which is why the auditor battery contains two compound mutations and why the one defensive line that no single mutation can expose is measured by pairing it with the mutation it guards rather than left to its own assertion. The domain gate of §4.1 is the same shape seen from the other side: its criterion is set membership over item ids, and reverting it to the equality of two counts changes nothing on any input the released pipeline can produce, because that pipeline already restricts the leaderboard to the shared leave-one-out domain. A revert of that kind is therefore not observable end to end, and we measure it where it is observable — by applying it to a copy of the auditor together with removing the upstream restriction, which turns a refusal into a verdict, and by checking that with the restriction removed the membership criterion

still refuses (`tests/test_revision_round4.py`). Re-running the battery also corrected a claim of its own: the auditor prints its own source digest as the first line of every report, and two of the compound judgments require two reports to be byte-identical, so that bookkeeping line made those judgments unsatisfiable, and the battery reported a working guard as disconfirmed. The comparison now excludes exactly that line and requires it to be present once, so that the widening does not go unnoticed and a missing digest does not pass. *Seed-dependent defects*: a defect is invisible for the particular random seed used in the test, so the suite must span seeds. *Assertions that stay true under the defect*: an assertion written against a constant that the defect moves to a different value under a different parameter setting passes when it should fail. *Fixtures too small to reach the assertion*: the test stays green because the branch it inspects is not executed at all, which is indistinguishable from a passing test unless one checks that the branch ran. A related discipline concerns statements about origins of defects. An early version of the latent-variable arm documented a symmetric fixed point as a consequence of uniform initialisation. Mutation measurement showed that three separate initialisation mutations left the output byte-identical, and that only a mutation of the update step reproduced the symptom. The comment now names the update step, and the ineffective mutations stay in the battery so that the correction remains checkable.

7 Artifacts and Reproduction

We release the auditor, the latent-variable arm, the exporters, the test suites, the mutation batteries, and the estimator sweep. Reproduction of every reading in this paper takes three routes, not two: one corpus is redistributed inside the release and needs no network, two judge-gold vote tables must be supplied out of band at their recorded digests, and the public-panel arm fetches its votes over the network and refuses to fold a summary when the fetch comes back short. The four human-gold annotation tasks are redistributed inside the release as flat files, and their SHA-256 digests, together with those of every generated report and the library versions used, are recorded in `evidence/manifest.json`. The two judge-gold vote tables are *named and digested but not redistributed*: their sizes, recorded in the manifest's own `datasets` section, put them outside the version-controlled tree, which lists them under `.gitignore` (`evidence/release_tree_check.json` records the clean-checkout measurement), so `make_manifest.py --check` on a fresh checkout exits non-zero and names them, which is the honest behaviour of a manifest that declines to certify a release whose inputs are not in front of it. The root digest is recomputed last and walks every section that carries a digest, nested gate reports included, and the published manifest counts 214 such entries; its `root_digest.per_section` field says how many came from each, so the coverage of the release identity is a number a reader can check rather than a word we chose. The registration covers the instruments and not only their outputs: each script that `reproduce.sh` and the arms it invokes execute is itself a registered entry in `evidence/manifest.json`, found by enumerating the chain's own invocations and asking, of each one, whether the manifest named it. Until that was checked, a reader holding the root digest could verify every report while the checker that certified those reports had been swapped, and nothing in the release identity would have said so. The enumeration is a judgement now rather than an author's one-off count: `make_manifest.py --check` re-derives the chain's invocations from the scripts themselves, refuses the release when one of them is not registered, and writes what it measured into the manifest's `chain_coverage` field, with the rule's own fixtures (a dropped registration, a second-layer script, a bare name that exists in two places, a reference that resolves to nothing, and a comment line that must not be counted) run by `make_manifest.py --selftest` in the reproduction chain. One assertion also runs against the rendered artifact rather than against source bytes: every long option the source typesets in a monospace context must extract from the PDF as an ASCII `--flag`, with its receipt in `evidence/pdf_text_layer.json` and `evidence/pdf_text_layer_anon.json` (`check_pdf_text_layer.py`). Under `T1` encoding with Latin Modern an unescaped double hyphen is set as an en dash even in a monospace argument, so the paper can print a command that no shell accepts while every source-level check stays green; the fixture receipt `evidence/pdf_text_layer_selftest.json` records that measurement case by case, including the case in which the correct spelling is also present, which a rule that only asked whether `--verify` appears would pass. One instance of exactly that shipped in this manuscript before the assertion existed, and it was found only by reading the produced text layer. A reader who supplies those two tables at the recorded digests has what the decision and question readings need to be rebuilt offline; a reader who works from the tracked tree alone does not have the inputs those two readings consume, and §8 counts that as a limit of this release rather than of the method. The exception is the public-panel reading of §5.7: its votes are cited,

not redistributed, because the release declares no licence, so that arm is produced by `rerun_panel.sh`, which fetches them, checks each file against `evidence/panel_votes_digests.json`, and refuses to fold a summary when the fetch returns a short or empty list; `rerun_panel.sh --verify` answers whether a shipped log was produced by the auditor now on disk, which is generated by a script rather than maintained by hand, so that `make_manifest.py --check` catches a stale digest before it can enter a release. Each reported figure is read from a report file, not transcribed, and seven of the paper's eight tables (Table 1, Table 3, Table 4, Table 5, Table 6, Table 7, Table 8) are regenerated from the carriers by one command, `make_tables.py`. The eighth, Table 2, is set by hand, and that is the table this paper got wrong once: its cells name verdicts rather than quantities, so we bind every verdict word in it, and in the abstract, to the report field that produced it with a second reverse check (`check_verdict_words.py`, registry `evidence/verdict-claims.json`), which is run over the source text and fails if a word is printed that no live carrier returns. A second file, `evidence/paper-ledger.json`, binds every number in the text to the artifact that produced it. Each entry carries the value, the file it came from, and either a literal match in that file or the name of a named formula that recomputes it; `check_manuscript_numbers.py` recomputes all of them and fails on any entry whose value does not follow from its stated source, so a number that drifts away from its evidence turns a check red rather than surviving quietly. The check is executed rather than asserted. We inject a mutated ledger for each class of defect — a changed value, a deleted and a renamed formula, an emptied provenance field, a missing source — and require all eleven mutants to turn the check red (`check_ledger_selftest.json`): a check that cannot fail is decoration. Writing that fixture found three real blind spots, which we closed rather than avoided. The latent-variable arm's four entries were bound by numeric token, so a value changed to another number that also occurred in the log passed; they now bind to a machine-readable summary of that log, and the checker requires the mutation arm to be field-bound at all, since a battery count whose own detector cannot be checked is the same kind of claim as a check that cannot fail. **Availability.** The release is a directory rather than a URL, identified by content instead of location: `evidence/manifest.json` publishes a root digest, a SHA-256 over the sorted path-and-digest pairs of every registered file, so two copies can establish that they hold the same release without either being hosted. A copy is supplied by the authors on request, and the request needs only the root digest, which identifies the version. Figures and tables are regenerated from these same carriers by `make_figures.py` and `make_tables.py`, so every value a figure shows is already bound in the ledger. Text drawn inside a figure reaches the produced artifact as text, and is checked there. The number gate reads the ledger and the report files its entries name; the verdict-word registry and the reverse-coverage scan locate their evidence in the source `main.tex`; and `check_pdf_text_layer.py` reads the rendered layer, where its seventh assertion takes the panel title and the axis label that `make_figures.py` writes today and requires each to appear as one contiguous label — a vector figure leaves its drawn text in that layer, so a panel shipped from a previous generator turns this step red, and the fixture that drives it red is the retired title itself. What no instrument of this release reads is a pixel: a panel that draws the right words over the wrong points is caught by regenerating the figure from its carriers and by the digest that pins the result, not by looking at it. The generator additionally refuses to draw when a panel title would name a different quantity than the axis it labels — the case that triggered that refusal was a panel titled after a threshold reference line while its axis measured the margin over the floor. The contents are the auditor, the latent-variable arm, the sweep, the test suites, the mutation batteries, the six datasets behind them (two judge-gold boards and the four human-gold annotation tasks of the third corpus), the manifest and the ledger, together with the exact commands that produced each report. `evidence/manifest.json` enumerates those files and records a SHA-256 for each, so the version this text describes is identified by content rather than by location, and `make_manifest.py --check` reports whether a copy on the reader's machine is that version. Three offline commands then re-derive what is claimed of it: `check_manuscript_numbers.py` recomputes each ledgered number from the report its entry names, `make_tables.py` regenerates every table in this paper from those reports rather than transcribing them, and `make_calibrated_summary.py` recomputes the calibrated arm's intervals, certification counts and δ-sensitivity of §5.8 from the per-board carrier. A fourth command packages the manuscript for arXiv: `make_arxiv_bundle.py` copies exactly the files the paper inputs — the source, the local style file, the expanded bibliography, seven table files and one figure — into a directory outside the release tree, compiles it with an empty `TEXINPUTS` in a scratch directory, and requires the page count and the author line of the result to match this build, so a package that silently omits a table it *inputs* fails here rather than in a moderation queue; its fixtures delete one of those tables and mis-state the expected page count, and require both to turn the check red. The test suites are self-contained and run offline. The

mutation batteries restore the implementation from a backup after each injection and verify the digest of the restored file, so a run that ends with an altered implementation fails rather than passing quietly. The estimator sweep of §5.12 runs in about ten minutes and writes both its per-seed numbers and its own console log, so the claim that the estimated span brackets the true span can be re-checked rather than trusted.

8 Limitations

We state twenty-two.

The decision corpus yields no verdict. Under the default tie rule the floor and the leaderboard do not share a denominator, because 28.6% of candidate items are generator disagreements that produce no label; under the rule that forces agreement, only two generators exist and the leave-one-out value degenerates into their agreement with themselves. Both outcomes are refusals rather than estimates (§5.1), and no threshold δ changes either.

The judges are the contestants. On all four annotator tasks the annotator set and the contestant set are identical, 25 of 25, so every judge grades its own work and the floor it helps construct is optimistic (§5.4).

The declared threshold is below the resolution of these corpora. Median minimum detectable differences of 4.55 to 15.63 points exceed $\delta = 2.0$ on all four tasks, giving power between 0.06 and 0.25 at that threshold. The verdicts we do report are not δ-tier readings and do not survive on that power: the four margins of §5.4 sit far above δ and are not floors at all (§4.2), the two corpora of §5.1 and §5.2 are refused by gates that return before δ is consulted, and the arms of §5.6 are decided at the equality branch, where a margin of zero clears nothing however wide the interval.

The above-median verdict does not control false positives under label noise. A gold-label permutation control on the human-gold chain — shuffling the external gold across items, which preserves each label multiset and destroys every alignment between a label and an item — measures what the verdict criterion returns on pure label noise: at $\delta = 2.0$ it still reports the leader above the median in 0.115 of 200 permutations on factuality, 0.885 on coherence, 0.000 on relevance and 0.35 on preference (`evidence/strict_review_followup/EC56_permutation_negative_control.json`). The rate tracks how unevenly the actors' answers fall across the label space: where the answers are nearly homogeneous, permuted gold rarely lifts the leader's margin past the median, and where they are not, label noise alone is enough to clear it. We state this as a boundary on the verdict rather than as a retraction of §5.4: the four margins and their intervals remain the substantive quantities, and the verdict attached to them is weak evidence exactly where this rate is high — an above-median reading on a task of this shape is worth trusting only to the extent its margin clears what permuted labels alone produce. On the m25 chain the within-item answer permutation passes trivially, and the pass is structural rather than reassuring: the leave-one-out identity pins the margin at zero for any assignment of answers to actors, so this control cannot discriminate on that chain — what discriminates there is the tie-rule and floor-choice sensitivity of §5.6 and §5.7. Nor is the rate a function of who sits in the pool: re-running the same control with the six Token Factory judges merged in — 25 judges growing to 31 on the 100-item domain the new judges share — moves it from 0.26 to 0.245 on factuality and from 0.96 to 0.965 on coherence, shifts of a point and a half at most, while shrinking the domain alone from each task's full item set to that same 100-item sample moves it from 0.0 to 0.255 on relevance and from 0.115 to 0.26 on factuality, so the rate tracks the answer distribution the domain presents rather than the pool's composition, and no enlargement of the pool repairs the coherence failure (`evidence/strict_review_followup/EC57j_pool_composition.json`).

Comparisons are corrected as a family, and the family is one task's 24 comparisons. Holm–Bonferroni rejects 23, 14, 18 and 3 of them on the four tasks, against 23, 21, 20 and 8 rejected by the same p-values read one at a time; the 70.8% is $1-(1-\alpha)^{24}$ with $\alpha = 0.05$, the uncorrected procedure's probability of at least one false rejection when all 24 nulls are true, not a claim about what share of our own rejections is wrong.

Two annotators are not enough. The decision-level corpus has exactly two gold generators, so its floor is well defined under majority but its annotator confusion structure is not estimable, and its leave-one-out value

is an agreement rate between the two rather than an independent reproduction rate; the latent-variable arm refuses this corpus by design, which means one of our three rulers is inapplicable to one of our two corpora.

Single corpora per claim. Each claim rests on one corpus. We report the design-choice sensitivity on two tiers of one corpus and on four tasks of another, and each claim rests on a single corpus.

The positive verdict has only fired in-silico. This is a method paper with negative results attached, and we say so rather than let the abstract imply otherwise. The READABLE arm of (3) is exercised in exactly one place here: the test suite's positive control (§6), whose generator pool and margin the fixture itself sets. The $m = 25$ judge-generated board of §5.6, whose gold we imposed, was built to be a second place and is not one: after the corrections described there, every decidable arm of it sits at a margin of exactly zero, which is the branch that reads UNREADABLE. The four human-gold reports of §5.4 carry `ABOVE_MEDIAN_ACTOR` in their released files; that word belongs to the second chain's own alphabet (§4.2), whose reference is the median of the ranked field rather than a leave-one-out floor, so it is not the arm of (3) either, and after the rename it can no longer be quoted as if it were. No real leaderboard of that shape returns a positive verdict in this paper; the public panel of §5.7 is a real released panel, and it is measured, but its contestants are its judges, so it exercises the `UNREADABLE` arm on third-party votes and still leaves the `READABLE` arm unfired outside the fixture. A preregistered calibrated variant of the floor (§5.8) does return certifications on this panel — 1,242 of 9,568 measurable resampled boards at the declared threshold, against none uncorrected — but the quantity it certifies is the variant's own debiased alignment, measured on the same self-grading pool with the external ruler entering through calibration and scoring only. A preregistered disjoint-field deployment (§5.9) does measure the configuration named below, on a released panel whose six judges and five contestants share no members: there the corrected rule returns `UNREADABLE` or `MARGINAL` on all 3,200 boards and certifies none, so the disjoint case is measured and its measurement is a refusal rather than a certification. What the definition below still excludes is a disjoint board that returns a positive verdict: on the one we measured the true margin is negative (-7.1 pts), because the judge pool's best member aligns with the external ruler better than the contestant field's best member does, and the rule correctly declines. §3 names the three published cases that bracket this: a public suite whose jury sits on its own board, a panel whose per-judge votes are released but whose ranked field is that same panel, and a panel built with the disjointness our rule requires whose vote table has not been released as of this check.

The reported denominators are the labelled ones. The decision corpus reports the 1,520 items that stay labelled once either generator is removed, out of 1,524; the question corpus reports the $12,406$ items on which both generators survive a leave-one-out reduction, out of $14,292$ (each generator's own reduced domain is printed as `loo_n` $= 12,903$ and $12,608$, which are scoring *rows* and not item counts, so the intersection is not their difference from $13,104$; the labelled sets are $12,305$ and $13,104$ items), and the annotator medians are computed on the intersection of annotator item sets, which on the preference task is 64 items out of 520. Where that set is small, the margin is wide and 16 of 24 systems are indistinguishable from the leader; we report both numbers rather than the margin alone.

The margin is a convention. The value $\delta = 2.0$ in (3) is not derived; it is a declared threshold, and a reader who prefers a different value can apply it to the margins we report without recomputation. A threshold sweep over $\delta \in [0, 20]$ and over the equality arm's tolerance across six magnitudes leaves every refusal in this paper where it is — the refused verdicts do not consult δ, and the m25 and panel margins are exactly 0.00 — but on the human-gold chain δ is load-bearing: each of the four above-median readings holds at every grid point below its own margin and flips once the grid passes it, between 4.5 and 13.6 points (`evidence/strict_review_followup/EC42_threshold_sweep.json`).

The floor is a construction, and the one we ship is one of them. The leave-one-out floor of (1) removes a single generator's ballot; a k-fold floor removes a fold of them at once. Recomputing the $m = 25$ construction at $k = 4$ — each fold holding out six or seven of the twenty-five ballots, the labelled set shrinking by between 2 and 41 items — moves three of the four task verdicts: factuality UNREADABLE $\to$ MARGINAL ($+0.62$ $[-3.11, +1.66]$), coherence UNREADABLE $\to$ READABLE ($+3.96$ $[+2.19, +5.84]$), relevance UNREADABLE $\to$ MARGINAL ($+0.51$ $[-0.41, +1.03]$), while preference keeps its domain refusal (`evidence/strict_review_followup/EC63_kfold_vs_loo.json`). The paired difference against the leave-one-out baseline is not reported: on this domain the leave-one-out floor loses items to ties, so the two margins

do not sit on one denominator; the k-fold margin itself is same-domain, and the comparison it supports is the one stated above. Substituting the median annotator for the leave-one-out floor raises the margin from 0.00 to +5.56, +19.33 and +6.69 points on factuality, coherence and relevance, each interval excluding zero (`evidence/strict_review_followup/EB_leader_vs_median.json`); on a board whose leader is itself a member of the gold-generating pool, that is the direction Proposition 4 predicts. The $m = 25$ readings therefore bound the verdict by the floor construction, and the construction we ship — leave-one-out, on the shared item set, with the tie rule declared and the interval printed beside every margin — is stated so that a reader can replace it.

The latent-variable estimator is naive and screened. It iterates the latent posterior and the confusion matrices, maximising the likelihood as in (11), rather than fitting a full latent variable model, and with two annotators and one label per item it is equivalent to majority vote. The model inherits the label-permutation non-identifiability of its class (32, 16): swapping the latent labels leaves the likelihood unchanged, so the likelihood cannot say which labelling is correct, and we anchor on agreement with majority vote instead. The same treatment of annotators as sources of noise is standard in crowdsourced annotation (29, 17, 22), where the marginal return of additional annotators is likewise reported as diminishing rather than absent. Its label permutation is unidentifiable, so each fit is started eight times and screened, and a value that only one starting point produces is reported as such: the coherence task's −7.60 points is the sole survivor of eight, with an anchor to majority vote of 0.836 against 0.988 to 0.994 elsewhere. Two of the four tasks were fitted after removing a label that no starting point could accommodate, so their floors describe a two-label space. We also re-ran the whole latent-variable comparison on case-level bootstrap resamples, and the signed differences carry into that resampling only where they were negative: on the two tasks with a negative Δ no replicate changes the sign, while on the two with a positive Δ—the two the identifiability screens already report as permutation-ambiguous—the sign fails to hold, the median replicate being of the opposite sign on preference (§5.10).

Self-grading and permutation screening are properties of this release, not of the method.

Absolute accuracies are identified but noisy. As described in §5.10, the estimated spans bracket the true span and run from 34.4 to 45.6 points across five seeds against a true span of 40, which is a spread and not a shift; the power probe shows both the bias and that spread falling with the item count, at a rate matching ordinary binomial sampling error, so the limitation is the sample size rather than a failure to identify the level. Ordering is the more robust reading, adjacent pairs aside. The sweep and the probe that establish this are reported in the release rather than asserted. **A refusal is a statement about the protocol, not about the models.** When the ruler returns a refusal it asserts that the quantity requested has no estimator under the conventions in force. It does not assert that the contestants are indistinguishable in ability: two systems may differ by any amount and still be unreadable by this instrument, and the reverse inference—that a refusal is evidence of no difference—is as unsupported as reading a margin as a measurement. **The resolution reported here is the instrument's, not the leaderboard's.** A corpus that cannot resolve a two-point difference is not thereby a bad benchmark; it is a benchmark this ruler cannot read at the margin it was published with. We state the minimum detectable difference for that reason, so the reader can tell an unreadable measurement from an uninformative one rather than taking our ceiling for a property of the data. **The construction assumes a checkable answer.** Every quantity here is an agreement with a label, so the rulers apply to tasks where a response can be scored right or wrong: classification, ranking against a reference, verifiable answer selection. On open-ended generation—style, creativity, harm—there is no gold the generator could be scored against under (1) in the first place, and nothing in this paper should be read as extending the floor to those cases.

The floor is measured before optimisation, and the rulers do not cover pairwise or post-training settings. Every quantity here is measured on annotations produced before any contestant is scored, so the rulers say nothing about what happens to a leaderboard once its gold becomes a training target. Goodhart's law and the machine-learning form of it (12, 14) describe the distortion that optimisation against a measure introduces, and (6) the same distortion in social measurement; none of that is measured here, and a corpus readable under our rulers could still be a measure that has stopped measuring. We state this because no floor measurement, including ours, licenses the reading that a readable leaderboard stays readable after it starts being optimised against. Nothing here is computed on pairs of responses, so the Arena-style boards

that rank by human preference under a Bradley–Terry model (8, 4), and the preference judgements that post-training consumes (26, 28), are outside the measurement even though they share the same noise-floor question. Whether a floor of this kind binds on a pairwise board is a measurement we do not make, and §4.5 states the boundary in full.

The latent-variable arm would misidentify under collusion. A group of annotators that produced correlated errors would resemble, to the estimator, a group of annotators agreeing with a truth; the confusion matrices absorb correlation into the latent class, and no statistic computed from the annotations can separate the two (16, 32). We report this as a limitation of the estimator rather than as a defect to be repaired, because repairing it requires a design the data does not have—annotators from outside the population that produced the contestants. **The $m \geq 3$ readings are overlapping pools, ours and somebody else's.** The verdicts of §5.6 are produced by redefining the gold of an existing human-gold set as the majority vote of its own 25 model judges, so generator pool and ranked field are the same 25 actors; §5.7 repeats the measurement on a panel whose votes were published by its own authors, which removes the suspicion that the zero margin is an artefact of *who* imposed the gold, but not the suspicion that it is an artefact of the pool grading itself, because on that panel too the ranked field is the generating pool; that overlap is a counted quantity rather than only a caveat, and its sign is fixed by the rule rather than observed — every one of the 25 systems scores higher against a gold it helped decide than against the same gold with its decisive items removed, which is what the plurality definition of decisiveness forces, since a decisive item is one the generator answers correctly by construction (§5.6 gives the argument and the carrier prints the decisive-item hit rate as a field); what the count sizes is the correct items each generator handed itself (medians +0.03 to +1.60 points, the four task medians of §5.6), and it does not test whether a judge prefers its own work, which would need a contrast that can come out negative and is not measured here —; a public leaderboard whose gold is produced by three or more independent generators would test the ruler harder, and we do not have one. The set we reuse is documented by its own authors as a reproducibility benchmark rather than a correctness one, and the 25 actors ranked there are its judges.

A serving layer sits below this measurement, and we did not measure it. §3 cites work showing that a model's own outputs can change with the serving configuration—batch size, GPU count and GPU type among them—independently of anything the model was asked to do, and that a production framework ships controls for it. Every gold label in this paper is a published vote and not a call we made, so no reading here can be affected by that layer; the limit falls on what the readings establish, not on their construction. What we did not do is the measurement that would put the two on one scale: nothing here estimates how much a judge's verdict moves when the stack underneath it is reconfigured, and the amplitude we print is the one that work reports for the model's own outputs, not for a judge. A benchmark that served its judges under varying load would carry a variance source this instrument does not see, and we can name the source without quantifying it, which is the boundary we mean to mark.

Two of the six datasets are not in the version-controlled tree. The two judge-gold vote tables are the inputs the decision and question readings are computed from. They are named and digested in the manifest's `datasets` section, and the auditor reads them from disk, but they are excluded from the tracked tree by `.gitignore`, so a reader who clones the release alone is missing the inputs those two arms consume, and `make_manifest.py --check` exits non-zero and names them. This is a property of this release, not of the method: everything else needed to reproduce the readings here, including the four human-gold tables, the auditor, the batteries and the ledger, is inside the tree. Closing it is a hosting decision (commit the two tables, move them to large-file storage, or publish them where the digests can be met), not a writing one.

Three gaps we did not close. First, the tie-resolution instrument of §5.2 measures 709 of the 799 items the gold leaves unlabelled and records 0 skipped for a missing probability, so the shipped carrier does not identify the remaining 90 or why they are outside its input; we report that as an open gap in our own instrument rather than as a property of the corpus. Second, Proposition 4 is a statement about a shape, not about the two boards we happened to have: it applies to any release that ranks the models whose votes make its gold, and we tested it on exactly two such releases because those are the only per-judge votes we could find. It says nothing about a panel disjoint from the field that returns a positive verdict; §5.9 measures one such disjoint panel and gets a refusal, at a negative true margin, so a disjoint certification remains unmeasured. Third, the four ABOVE_MEDIAN_ACTOR readings of §5.4 rest on margins between the

leader and the median annotator; a case-clustered paired interval for that contrast is in the release carrier (`evidence/ds_median_interval.json`), and three of the four intervals exclude zero, while the preference task's does not on 16 cases, so that margin stays unresolved at this resolution.

9 Conclusion

For a leaderboard whose gold labels are the plurality aggregation of (1) and whose gold-generating pool is a strict subset of the ranked field, this audit emits a readable verdict only when the leader clears the producer floor; that floor is determined by choices which are usually unstated. Both conditions are load-bearing, and none of the five readings this paper reports satisfies them together: the two two-generator corpora fail the first, the construction and the released panel fail the second, and the human-gold set has no model-generated gold at all. On the corpora we measured, those choices change what the verdict is worth rather than which verdict it is: both gold definitions leave the question corpus refused rather than decided, the majority reading keeps a margin whose interval excludes zero while the comparison that would have to carry it is refused, and on the decision corpus the point difference clears the declared threshold while its interval contains zero — which is why the instrument reports a refusal there too. The latent-variable alternative moves the floor in opposite directions on different tasks. The floor and the leaderboard must be measured on the same items for that margin to mean anything, and a margin without an interval is a number rather than a measurement; in our measurements the item sets the labels support are smaller than the corpora and are named, and every reported margin carries the interval that says how little it settles. We release the measurement, three rulers with an explicit refusal when the annotations fail to support the quantity requested, a construction on which the same instrument pins every arm at a margin of zero, and a mutation battery that measures how much of our own test suite can fail. It has now been pointed at one public leaderboard of the shape it was built for: a preregistered disjoint-field deployment (§5.9) whose six judges and five contestants share no members returns a refusal on all 3,200 boards and certifies none, because the judge pool's best member aligns with the external ruler better than the contestant field's best member does. So the READABLE arm of (3) still fires only on the test suite's positive control, and a positive wild reading is the next measurement, not a finding of this paper. A preregistered calibrated variant of the floor does return certifications on the released panel — the corrected rule certifies on 1,242 of 9,568 measurable resampled boards where the uncorrected rule certifies on none, and on the configurations that leave a true external margin the false-certification rate is between 0.000 and 0.069 — but the certified quantity is the variant's own debiased alignment (§5.8), on a pool whose contestants are its judges.

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A Withdrawn figures

Several numbers in earlier versions of this paper were produced before the corrections described above and are withdrawn. We keep them here, with the defect that produced each, because a number that disappears without a record is harder to audit than one that is retracted in place. Three of the entries below come with the bytes that produced them. The $m = 25$ readings were produced by the gold rule of the previous round, and the reports as that rule left them are archived in the release as `evidence/pre-tiefix__m25_{summary,factuality,coherence,relevance,preference}.json`; the question-corpus mean-probability readings were produced before that gold was fixed, and the reports as the defect left them are archived as `evidence/pre-meanprobfix__jev_question_{both,lexicographic}.json`. Both archives let a reader re-read a withdrawn verdict and its interval from bytes rather than from our prose. The third is the forced-rule reading of §5.1, whose number did not move and which carried the wrong verdict word, so the live report already shows it. The remaining entries have no archived carrier, because the defect was in what we wrote about a run rather than in the run; for those, the section named at the end of each entry is the replacement, and each now states what that section reports instead.

- **Decision corpus,** READABLE **by** +2.96 **points.** The first version reported a leave-one-out floor of 70.23%, a leader at 73.19% and a verdict of READABLE. The tie rule behind it resolved 436 of 1,520 items by sorting the two answers as strings, an arbitrary rule that also moved the leader. Withdrawn in §5.1. *Now reads:* no verdict on either rule, with the two measurements stated separately and not subtracted from one another.

- **Decision corpus, leader margin interval** $[-0.14, +6.27]$**.** Reported as a verdict-supporting interval on two terms measured over different item sets. Withdrawn in §5.1. *Now reads:* the interval is printed on the refused row as reference only, and does not carry a verdict.

- **Decision corpus, a δ-sensitivity claim.** An earlier version stated that at $\delta = 0.0$ the corpus would remain READABLE. The claim was not recomputed against the corpus and stood in tension with the interval the same section reported. *Now reads:* no δ-sensitivity analysis is reported for this corpus, and the reason is given rather than omitted: neither rule produces a verdict at any δ.

- **Latent floor deltas** $+1.40$ **and** -6.25 **points.** Produced by a single deterministic initialisation of an estimator whose seed argument was accepted and ignored, so every seed ran the same trajectory. Withdrawn in §5.10. *Now reads:* the four signed differences of Table 8, each produced by a fit whose seed argument is consumed and whose starting points are screened.

- **Estimator compression "from** 40 **points to between** 22 **and** 27**".** That range came from a 40- to 60-item variant in which several annotators are not identifiable at all and report no estimate; a missing estimate is not a compressed estimate. Withdrawn in §5.12. *Now reads:* the estimated span brackets the truth rather than compressing it, and the sweep reports the spread it observed.

- **Sweep over ten seeds with a Spearman floor of** 0.943 **and deviations of** 1.9 **to** 0.5 **points.** Produced by the same seed-argument defect as the deltas above: ten seeds ran one trajectory, so the reported spread was zero by construction. Withdrawn in §5.12. *Now reads:* the five-seed sweep of §5.12, with the ordering recovered in all five at the top rank and the full ordering in four.

- **Preference-task margin of** 5.77 **points.** Computed before the four human-gold margins were placed on a common item set. Withdrawn in §5.4. *Now reads:* the fourth margin as a different and better-defined quantity, computed on the intersection of the annotator item sets.

- **Auditor mutation count of** 23 **with one survivor.** The one uncaught mutation was a defensive line that no single mutation can expose; the battery now measures it by pairing it with the mutation it guards.

- **Question corpus** UNREADABLE **under the forced tie rule.** Reported at -1.57 points; withdrawn as UNDECIDABLE_DOMAIN_MISMATCH (§5.2).

- **The four verdicts of the** $m = 25$ **construction, §5.6, as first reported.** The registered readings were UNREADABLE at -0.0033 on factuality, MARGINAL at $+1.6037$ $[0.5276, 2.9412]$ on coherence, MARGINAL at $+0.1023$ on relevance and a refused -0.0048 on preference. Two defects produced them, both in the gold rule rather than in the data. The tie test was $\max(c) \cdot 2 = \sum(c)$, which recognises only an exact two-way split: on 19 items whose largest vote count was shared with a third answer holding the remainder, the gold fell to the lexicographic order of the answer strings — the very rule this paper refuses to adjudicate with — while the same test voided 5 items that do have a unique plurality winner (4/2/2). That defect moved the gold, and with it both terms of the margin: coherence's leader `kimi-k3` went from 82.5737% under the old table to 81.1236% under the corrected one, and the floor from 80.97% to the same 81.1236%, so the two moves add up to exactly the 1.6037 that disappeared. The second defect is separate and small in that reading: the per-generator leave-one-out values were rounded to two decimals before the maximum was taken, while the leaderboard was carried at four, so every printed margin mixed precisions: the difference it reported could sit as far as half a unit of the floor's last displayed place away from the unrounded one. That half-unit is immaterial against $+1.6037$, but it becomes decisive precisely in the case the corrected gold produces, where the leader and the floor are the same number and a margin of zero must be read as zero rather than as the residue of a rounding. With both corrected — a gold label exists iff the largest vote count has a unique holder, and ceilings are compared at full precision — every arm of that construction sits at a margin of exactly 0.00: see §5.6 and `evidence/m25_summary.json`.

- **Question corpus, mean-probability gold reading** 8.97% **with `jev-1.13.0` first.** The mean-probability column of Table 4, the spans of §5.3 computed under that gold, and the rank-flip count

were all produced by a gold label truncated to its first character: the implementation took the maximum of a mapping from answers to weights and then indexed the result twice, and since the first index already returns the answer string, the second returned its first character. Answers that differ anywhere then share a label with several others, so on the 799 items where the two generators disagree the gold frequently names a string that is neither generator's answer. The board that came out of it put a leader at 8.97% on 11,897 scoring rows against a floor of 89.48% — a margin of −80.51 points with interval [−81.36, −79.59] — with six systems changing rank, and that is what §5.2 earlier reported as the two conventions disagreeing about the leader outright. Two facts make this entry worth keeping rather than quietly regenerating. The first is that the defect is invisible from the shape of the output: the report parsed, the fields were there, and the arithmetic inside it was correct, so a reader who checked the margin against the two printed numbers would have found them consistent. The second is why it survived two rounds of review: the arm that computes this corpus wrote a file under one name while every consumer — the number gate, the table generator, the figure generator, the manifest — read a differently named file that no command wrote, so the tool was fixed and the shipped readings stayed in the generation that carried the bug. The fix is one index removed in `audit.py`; its regression fixture is a two-judge toy whose gold answers are longer than one character, red before the fix and green after, and it is case 1 of the suite in `selftest.py`; the archived reports above let a reader re-read 8.97%, −80.51 and the six flips from bytes rather than from this paragraph.

- **"**MARGINAL **includes the case where the leader equals the floor."** The declared rule is a strict inequality, $\max_s \mathrm{acc}_c(s) > \mathrm{floor}_c + \delta$; a margin of zero is not a clearance by a small amount but no clearance at all. The chain tested ceiling > top first and fell through to top − ceiling < δ otherwise, so equality printed MARGINAL. This only became visible because the corrected tie rule made equality the common case.

Two of the figures above survive only in this list. The 1.9-point sweep bias and the pre-alignment 5.77-point margin came from generations whose reports are not in the release, and rather than leave them bound to files that no longer contain them, the ledger marks both `recoverable: false` and the reverse-coverage scan prints them as declared-unrecoverable on each run; a withdrawn number that is not re-readable is still a number the reader should be able to see was withdrawn. The other withdrawn figures are re-read on each run either from the live reports, where the value still exists, or from the archived copies of the previous generation that the release carries (`evidence/archived_v2_*.json`) at a named field, and the scan's receipt records which route each one took.